



A+ LAB SERIES V5

Lab 17: Security and Privacy

Document Version: **2026-01-02**

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	3.5: Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards.
CompTIA A+ (220-1202) Exam Objectives	2.1: Summarize various security measures and their purposes. 2.2: Given a scenario, configure and apply basic Microsoft Windows OS security settings. 2.4: Summarize types of malware and tools/methods for detection, removal, and prevention.

Contents

Introduction	3
Objective	3
Lab Topology	4
Lab Settings	5
1 Windows Defender Security Center	6
1.1 Scanning your System for Malware in Win10-OS	6
1.2 Viewing the Health and Performance Report with Win10-OS	11
1.3 Managing Network Security in Win10-OS	13
1.4 Using the Allowed Apps Feature in Win10-OS	15
1.5 Adding Apps to the Allowed List in Win10-OS	19
1.6 Removing Apps from the Allowed List in Win10-OS	21
1.7 Windows Firewall Advanced Security in Win10-OS	22
1.8 Creating New Rules with Windows Firewall in Win10-OS	29
2 Windows Encryption	36
2.1 Encrypting File System in Win11-OS	36
2.2 Enabling BitLocker without TPM in Win10-OS	47
2.3 Disabling BitLocker with PowerShell in Win10-OS	55

Introduction

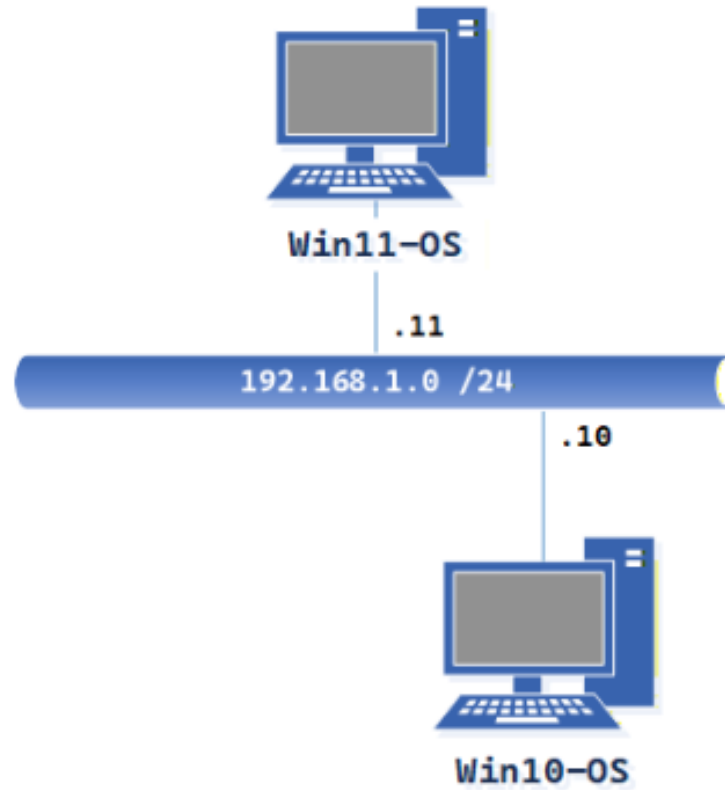
In this lab, you will explore essential tasks to bolster the security and integrity of your Windows system. Throughout this session, you will complete a series of practical exercises to equip you with the skills to protect your PC against various threats and optimize its performance.

Objective

In this lab, we will be performing the following tasks:

- Scan your system for malware using the Windows Defender Security Center.
- Use Windows Defender Security Center to review device performance and health.
- Use Windows Defender Security Center to review Firewall & network protection.
- Navigate the Control Panel to access Windows Defender Firewall to allow specific apps. Add and remove apps from the Allowed list.
- Use Windows Firewall Advanced Security to modify the ICMP rules and create new rules.
- Protect your PC against malicious code
- Enable and use EFS to encrypt files.
- Explore and enable BitLocker without TPM. Disable BitLocker with PowerShell.

Lab Topology



Lab Settings

The information in the table below is needed to complete the lab. The task sections below provide details on how to use this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Windows Defender Security Center

1.1 Scanning your System for Malware in Win10-OS

Windows Security Center, now known as **Windows Security**, is a unified platform designed to protect your Windows 10 and 11 devices. It offers a comprehensive suite of security features, including antivirus, firewall, and web protection. Let's delve deeper into its key components:

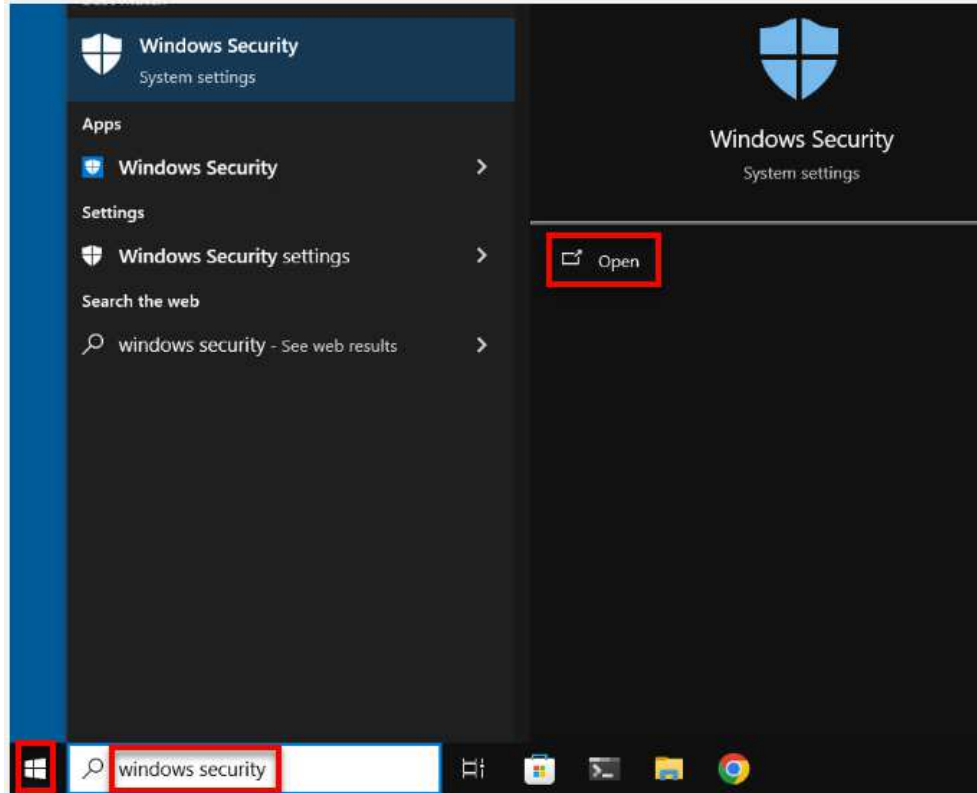
- *Virus & threat protection* – includes Windows Security Antivirus settings and monitoring features.
- *Device performance & health* – allows the verification of information about drivers, storage space, Windows Update issues, and a mechanism for fixing errors on the PC.
- *Firewall & network protection* – monitor the status of network connections and access to firewall settings, including Windows Defender Firewall.
- *App & browser control* – helps protect the system and data from malicious code hidden in applications, files, and websites.
- *Family options* – includes access to parental controls and a kid-safe online experience using a Microsoft account.

1. Open a console connection to the **Win10-OS** system.



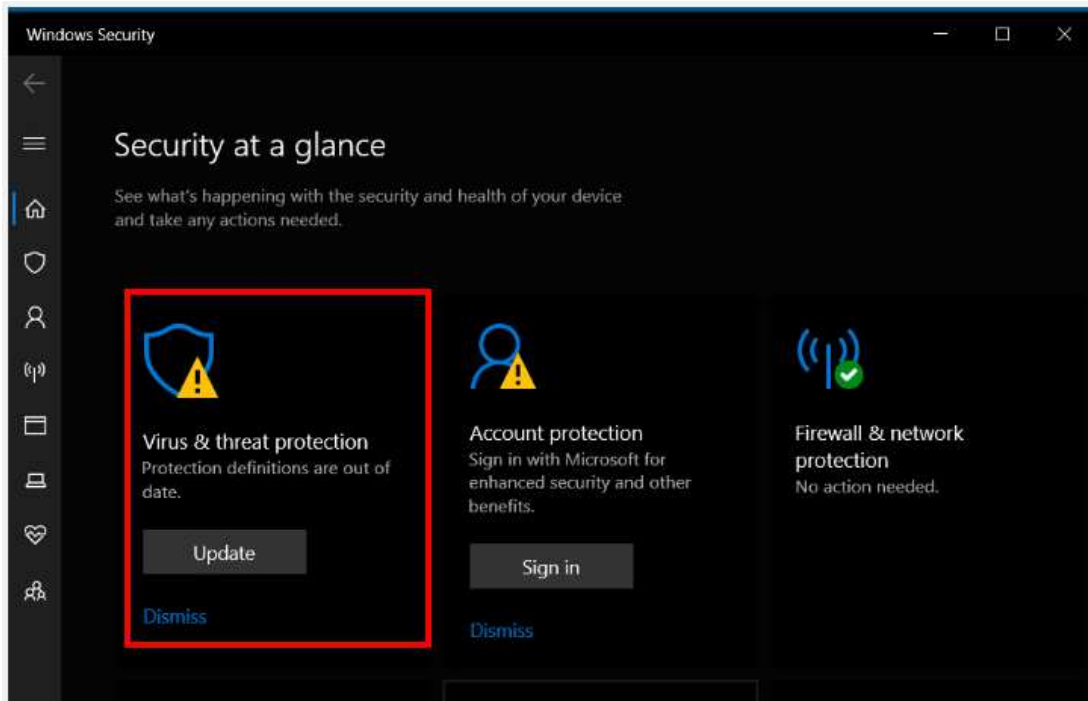
To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

- Click on the **Windows Search** icon and type **Windows Security**. From the search results, click **Open**.



Windows Security Center in Windows 10 acts as a dashboard for all the security features, including third-party security, providing an overview of the risks on the machine. It is designed to simplify and centralize security settings of Windows.

4. Notice the security center window appears. By default, it opens to the Home dashboard, where you can view the status of all security features and alerts, and take any actions needed on the system. The dashboard will also display antivirus information, such as when the last scan was performed and when the last definition was updated. There are diverse types of scans that can be performed manually. Click on the **Virus & threat protection** icon.



When all is well, you will see the green checkmark overlay. If an issue requires your attention, you will see a yellow exclamation mark overlay.

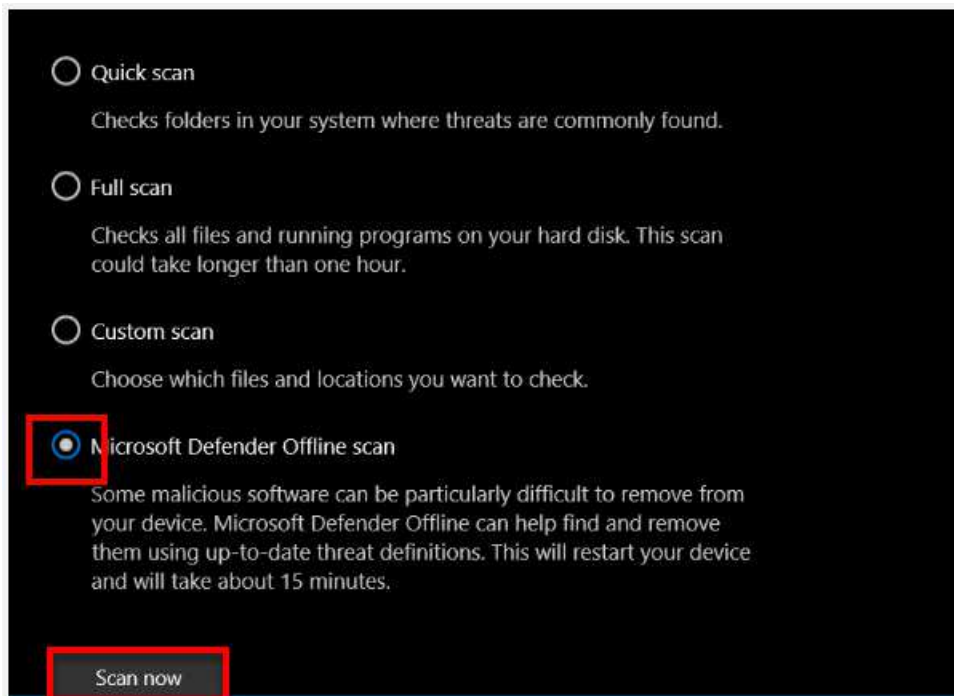
5. Notice that a *Quick Scan* option is available here (if not immediately available, wait a few seconds). As the name indicates, this is a fast scan that targets only the parts of the system where malware is known to hide. Click on the **Scan options** link.



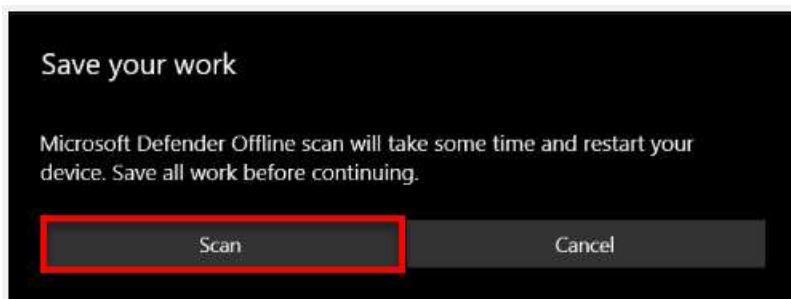
Notice several types of scans are available here:

- **Quick Scan** – Checks folders where threats are commonly found.
- **Full scan** - This scan will scan your entire hard drive, including all files and programs.
- **Custom scan** - This scan allows you to choose the locations and files which you would like to scan. There are two ways that a custom scan can be run. One method is to select the radio button next to *Custom scan* and then click the *Scan now* button. A second method is to right-click a drive, folder, or file and select the *Scan with Windows Defender* option from the context menu to perform a custom scan.
- **Microsoft Defender Offline scan** - Depending on the type of malware infecting the system, the antivirus software may not be able to remove it while *Windows* is running. It may be necessary to run an offline scan.

6. Select the radio button for **Windows Defender Offline scan** and click the **Scan now** button.



7. Click **Scan** to continue.



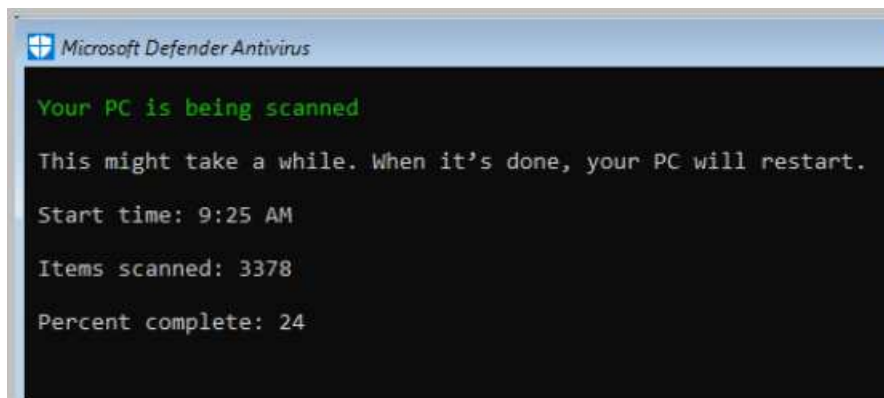
8. When prompted with User Account Control, click **Yes**.



9. Notice that *Windows* will automatically sign you out and reboot the system to begin the scanning process. It will boot using the *Microsoft Defender Antivirus (offline scan)* environment. The latest definition update will be installed (if an Internet connection is available). Since this lab environment has no Internet connection, it will skip the update process and begin scanning. After the scan is complete, the system will log in automatically to the sysadmin account. As the scan runs, you will see a notification that Microsoft Defender Antivirus is loading.



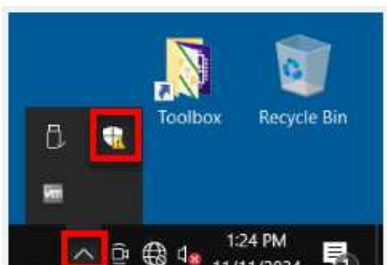
10. Once loaded, the scan will start as shown in the screenshot below.



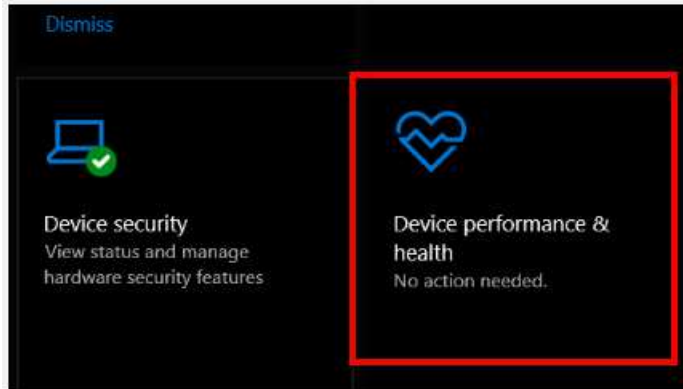
11. After the scan is complete, *Win10-OS* will automatically boot and log in with the sysadmin account. Leave the *Win10-OS* window open to continue with the next task.

1.2 Viewing the Health and Performance Report with Win10-OS

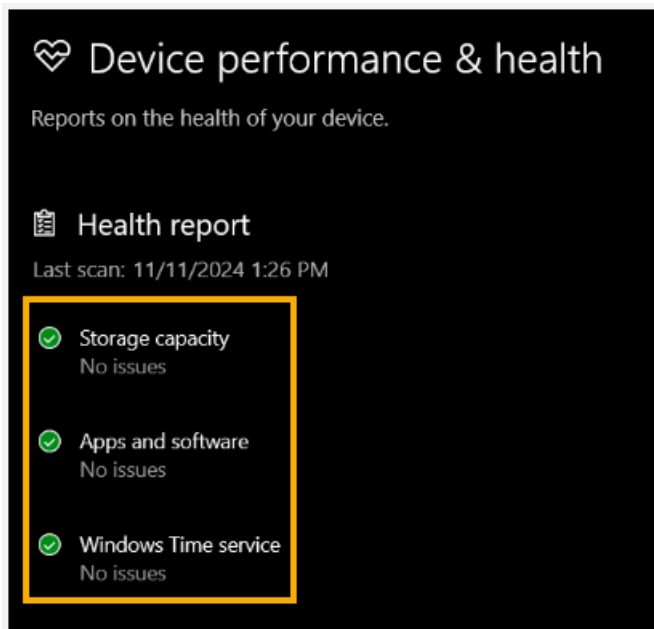
1. Click on the **Show hidden icons** arrow in the *System Tray*, followed by clicking on the **Windows Security** icon (if it isn't visible right away, wait a couple of seconds for the icon to appear).



2. In the *Windows Security Center* window, click on **Device performance & health**.



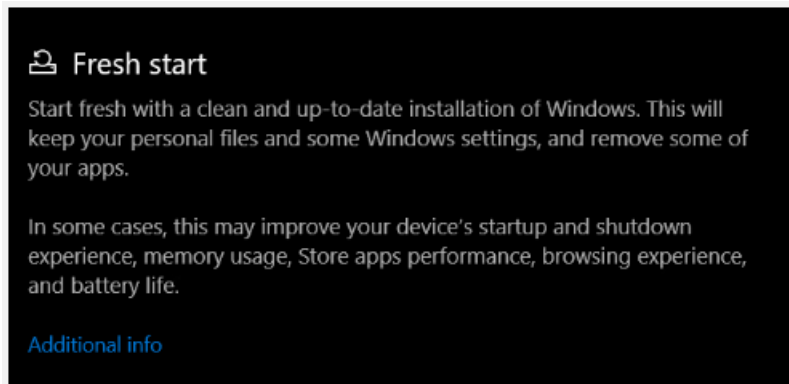
3. A colored symbol indicates the status of each of the options: green is good, yellow is a recommendation, and red is a warning.



Health report is the header for the area which tells you if there are any issues impacting the device health.

- *Storage capacity* - Checks if your device has enough space to update Windows.
- *Apps and software* - Checks the health of installed apps and running software.
- *Windows Time service* - is a crucial component of Windows operating systems, responsible for synchronizing clocks

4. If a system is having issues that are difficult to resolve or it needs to be refreshed, the *Fresh start* feature can be used to perform a clean reinstallation and update of *Windows* while keeping your data and most *Windows* settings intact. If the *Win10-OS* system had internet access, you could click *Additional Info*, which would take you to *Microsoft.com* for the steps to check your device performance and health in Windows Security.

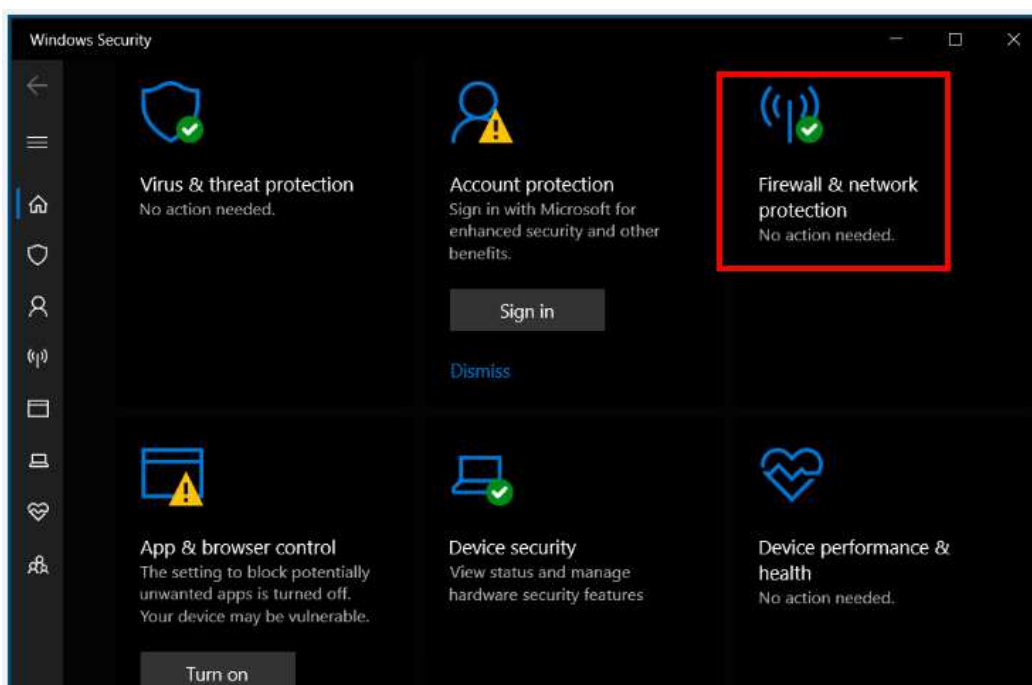


5. Click the **back arrow** to go back to the main *Windows Security Center* dashboard.
6. Leave the *Win10-OS* window open to continue with the next task.

1.3 Managing Network Security in Win10-OS

Windows Security Center also provides access to the area for monitoring and managing *Windows Firewall & Network Protection* settings. Firewall settings allow you to control inbound and outbound traffic, *TCP* ports, and content. Using *Advanced Windows Firewall* rules, you can customize rules that apply to specific ports, protocols, services, and all kinds of advanced criteria.

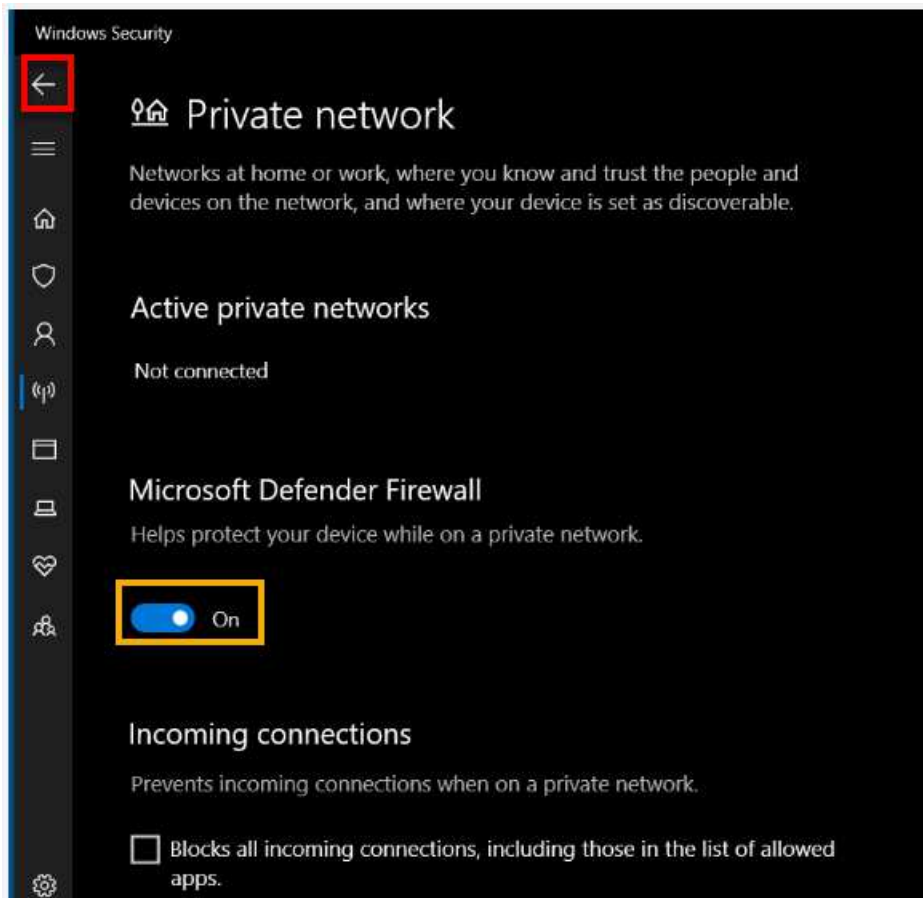
1. In the *Windows Defender* window, click on **Firewall & network protection**.



- This opens the page where you can view network profiles. Click on **Private network**.



- Review the available options on the page. Notice that one of your options here is to turn on/off the firewall. Click the **back arrow**.





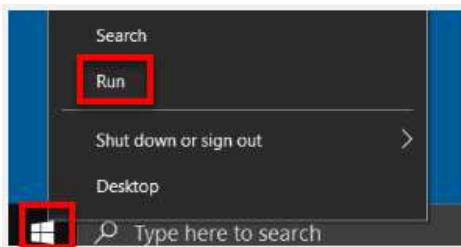
Users must have an administrator account to customize the rules in *Windows Firewall*. *Firewall & network protection* includes access to several other advanced firewall settings such as Allow an app through firewall, Network and Internet troubleshooter, and others.

4. Close the **Windows Security Center** window.
5. Leave the *Win10-OS* window open to continue with the next task.

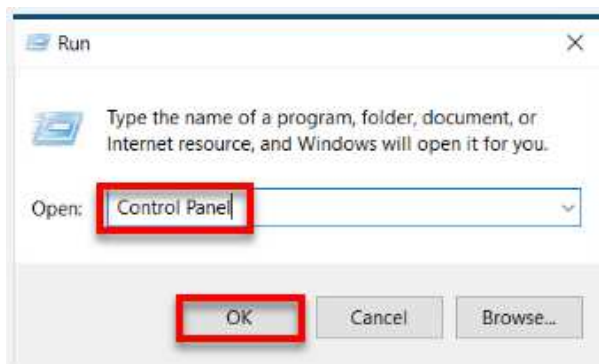
1.4 Using the Allowed Apps Feature in Win10-OS

Now, you will launch *Windows Defender Firewall* using the run command. This method is available in all versions of *Windows* being used in the lab.

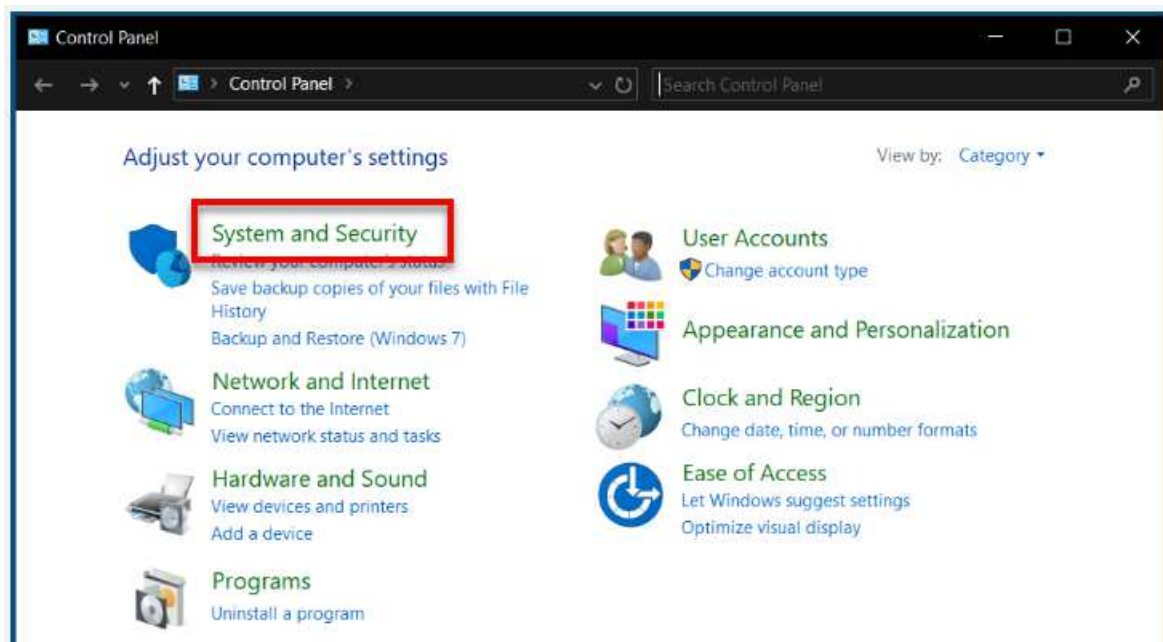
1. Right-click the **Start** button and click **Run**.



2. Type **Control Panel** in the text field and click **OK**.



3. In the *Control Panel* window, click on **System and Security**.



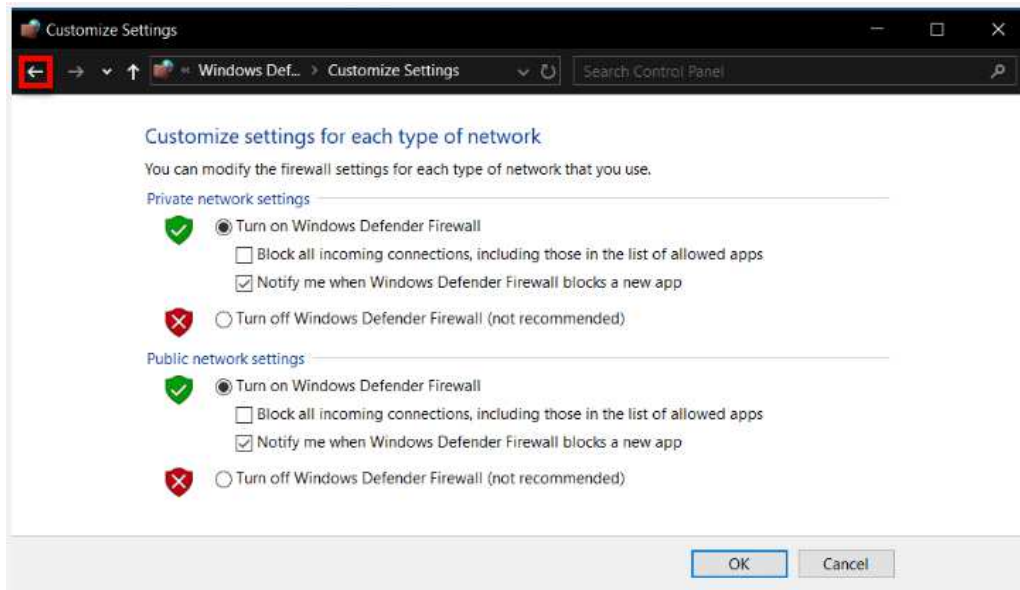
4. Click on **Windows Defender Firewall**.



5. To enable or disable the *Windows Defender Firewall*, notice in the left column the **Turn Windows Defender Firewall on or off** link, click it.



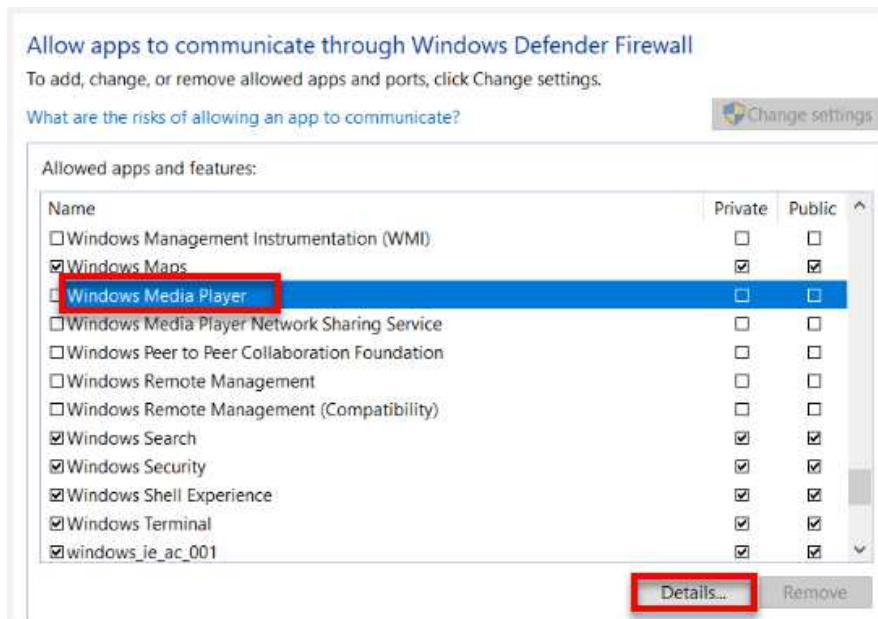
6. In the Windows Defender Firewall Customize Settings window, you can customize how the firewall operates for each network profile. Click the **back arrow** to continue.



7. In the left column, click the **Allow an app or feature through Windows Defender Firewall** link. This feature allows any user with an administrator account to change the list of rules and exceptions applied directly to applications.



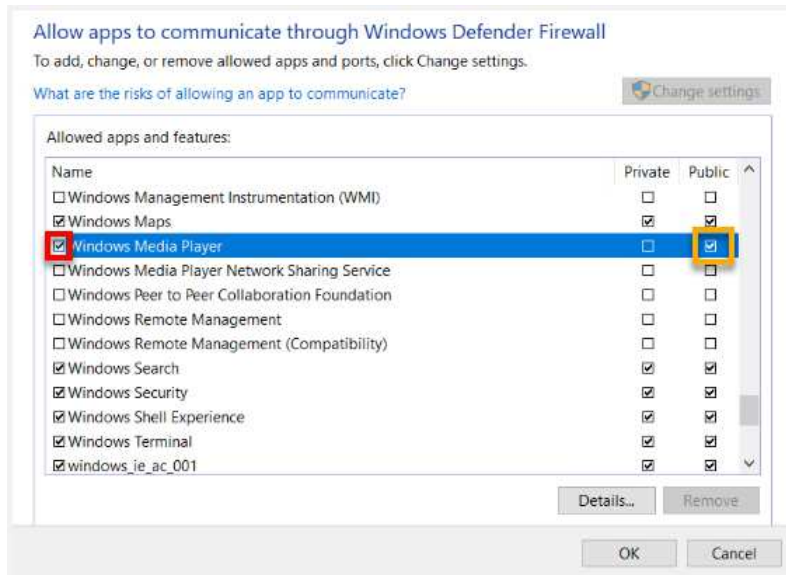
8. Notice apps, programs, or features in the list have checkmarks on the left side of their name to show the rule is enabled and used by the *Windows Defender Firewall* to allow access. The *Private* and *Public* columns on the right also have checkmarks to indicate whether the rule is active when the device is connected to that particular network profile. Scroll down in the list, and select **Windows Media Player**, followed by clicking the **Details** button.



9. The details pane gives a bit more information about the application or feature relevance to the firewall viewpoint. Click **OK**.



10. Currently, *Windows Media Player* is configured as a “disallowed” application. **Check** the checkbox to the left of the entry. Notice that the checkmark in the *Public* column will automatically check itself. This will allow its access only in the public network profile.

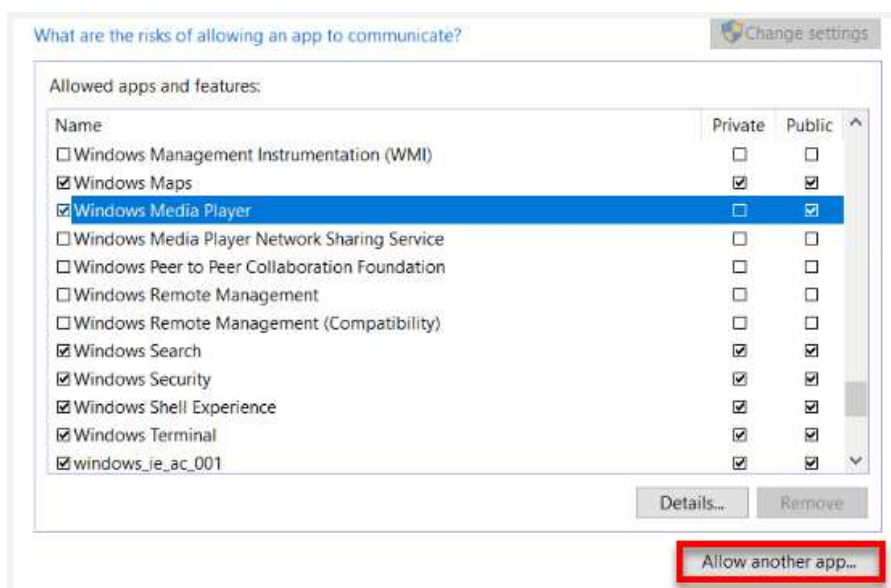


11. Leave the *Allowed apps and features* window open to continue with the next task.

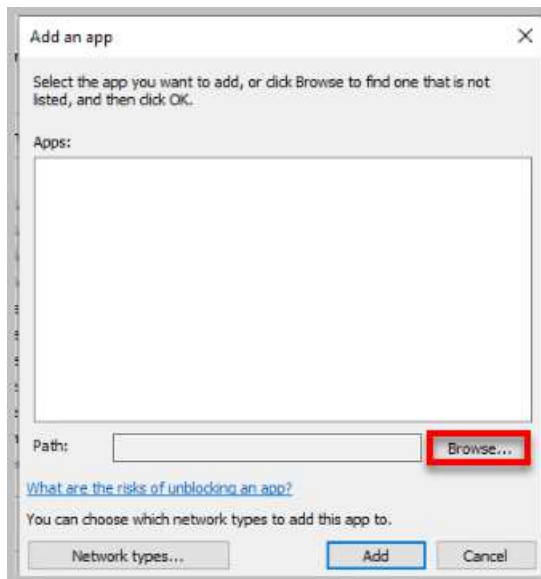
1.5 Adding Apps to the Allowed List in Win10-OS

If a program is not in the “allowed” list but needs to communicate with the network, it can be added. If there is a program that should not communicate with the network, you can remove it from the “allowed” list. This is done using the "Allow an app or feature through Windows Firewall" link.

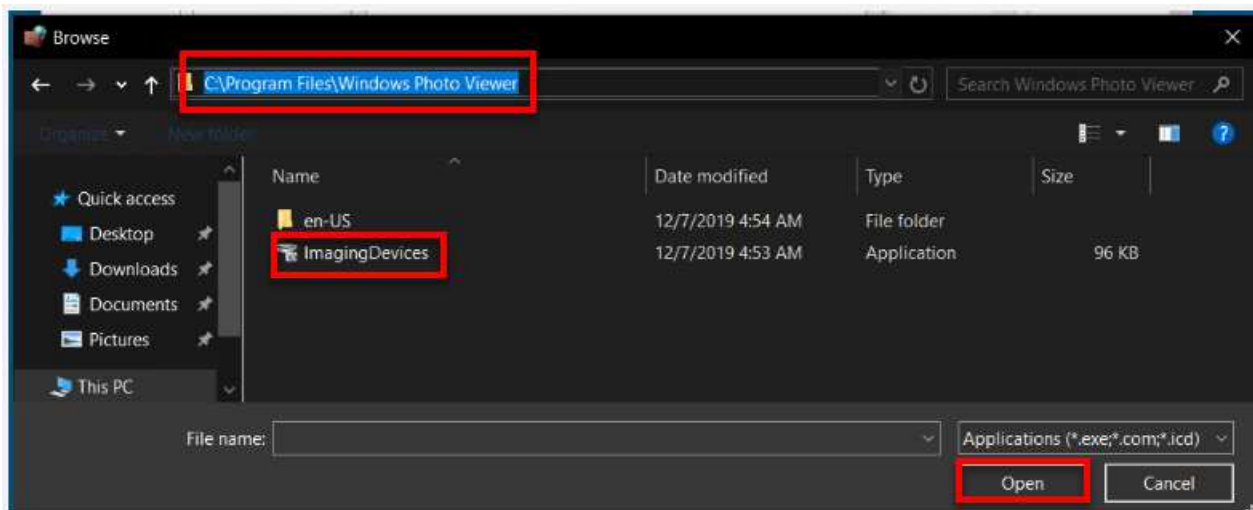
1. In the *Allowed apps* window, click on the **Allow another app** button.



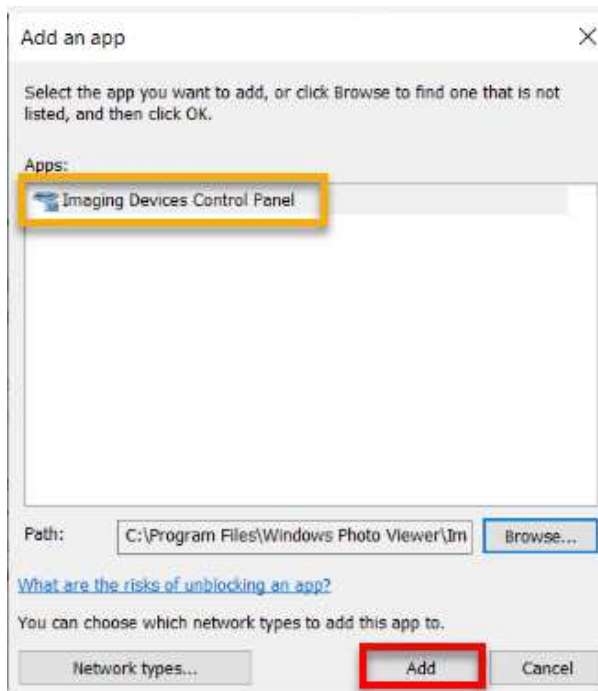
- Click the **Browse** button.



- In the *Browse* window, navigate to the **C:\Program Files\Windows Photo Viewer** directory and select **ImagingDevices.exe**. Click **Open**.



- Back on the *Add an app* window, confirm that the *Imaging Devices Control Panel* app appears and click **Add**.



- Locate the *Imaging Devices Console* app in the list. Notice that the app is enabled in the *Public* network profile.



- Leave the *Win10-OS* window open to continue with the next task.

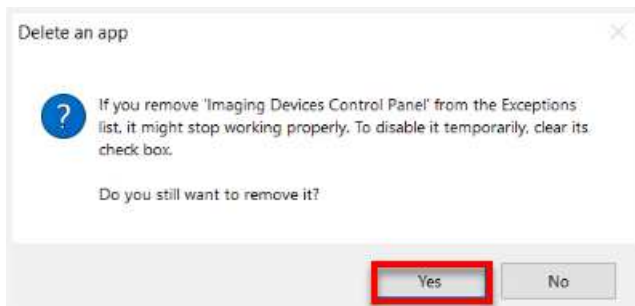
1.6 Removing Apps from the Allowed List in Win10-OS

When removing a program from the list, it will be blocked, so an approval request for network access will be made each time the program is used.

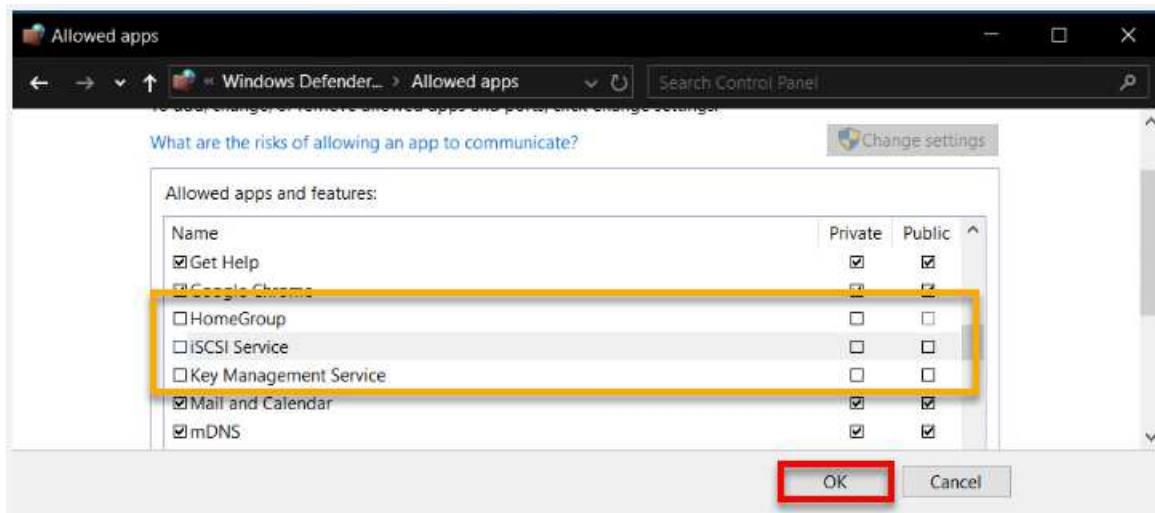
1. Select **Imaging Devices Console** followed by clicking on the **Remove** button.



2. When prompted, click **Yes** to continue.



3. Notice *Imaging Devices Console* no longer appears in the list. Click **OK** to return to the Windows Defender Firewall page.



4. Leave the *Win10-OS* window open to continue with the next task.

1.7 Windows Firewall Advanced Security in Win10-OS

Windows Firewall with Advanced Security is a management snap-in that controls all rules at a very detailed level. It is where you can edit all rules and exceptions at a very granular level.

In this exercise, you will modify the *ICMP* rules to allow the *Win10-OS* and *Win11-OS* systems to ping each other. When *Windows Firewall* is enabled with default settings, you cannot successfully ping from another device to test connectivity.



Three important types of rules of *Windows Defender Firewall*. *Inbound* rules apply to traffic that is coming from the network or the Internet to your *Windows* computer or device. *Outbound* rules apply to traffic that is originating from your computer and going to the network and the Internet. *Connection security* rules secure the traffic between two specific computers while it crosses the network.

1. In the *Windows Defender Firewall* window, on the left pane, click on **Advanced settings**.

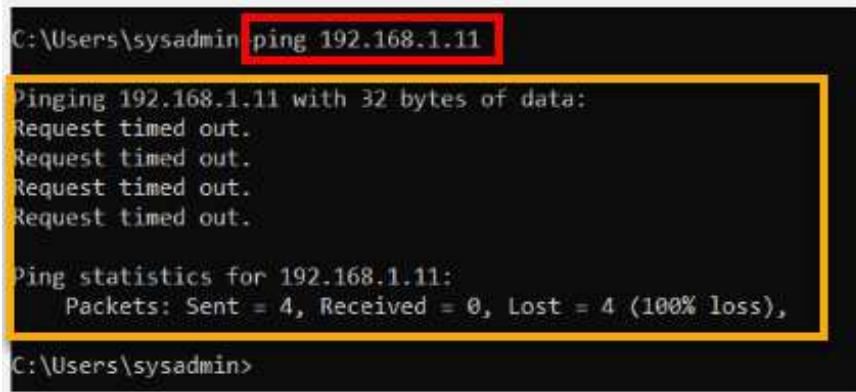


2. Click the **Windows Search** icon in the taskbar and type `cmd` followed by pressing the **Enter** key.



3. In the command prompt window, type the command below, followed by pressing the **Enter** key. Notice that the *Win10-OS* system fails to successfully ping the *Win11-OS* system.

```
ping 192.168.1.11
```



```
C:\Users\sysadmin> ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

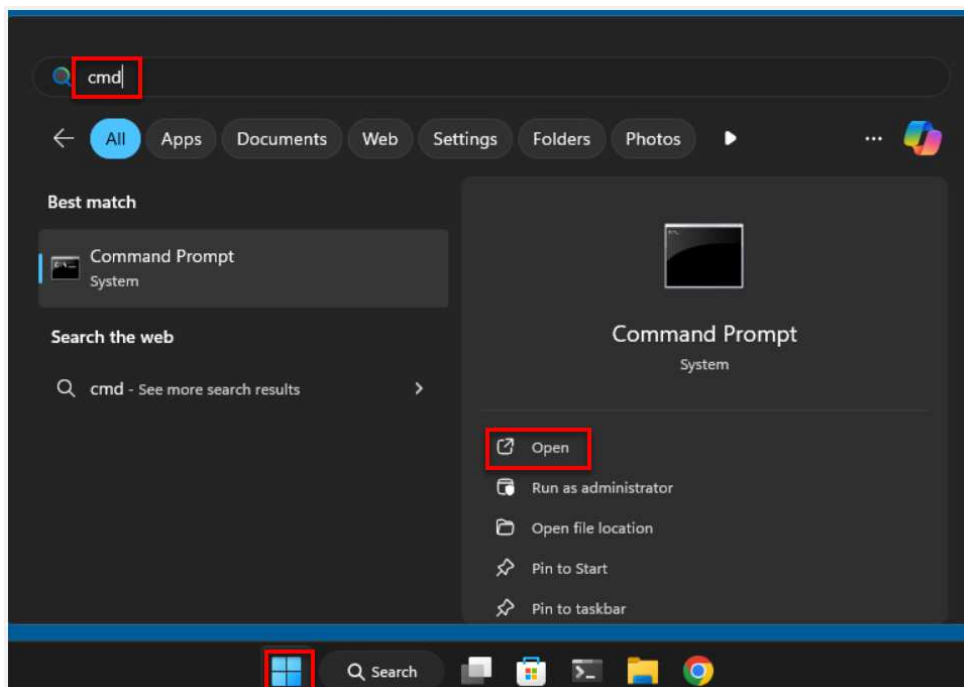
Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\sysadmin>
```

4. Connect to the **Win11-OS** console.



5. Click the **Start** button and type **cmd**. Click **Open**.



6. In the command prompt window, type the command below, followed by pressing the **Enter** key. Notice that the *Win11-OS* system fails to ping the *Win10-OS* system successfully.

```
ping 192.168.1.10
```

```
C:\Users\sysadmin>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

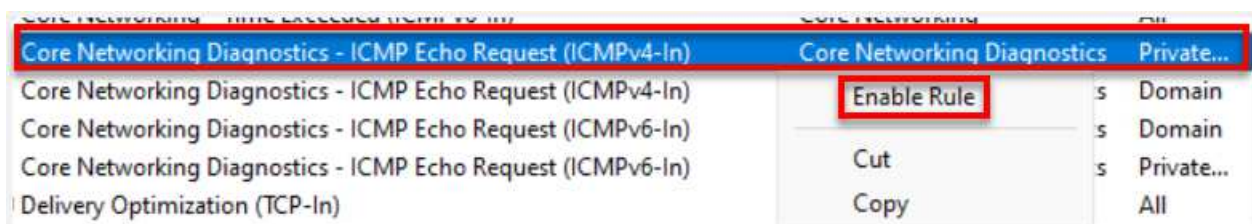
Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\sysadmin>
```

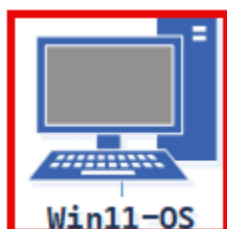
7. Change focus to the **Win10-OS** window.
8. While accessing *Win10-OS*, focus on the **Windows Defender Firewall with Advanced Security** window.
9. In the left pane, you will see several types of rules that can be applied. Click on **Inbound Rules** located in the left pane.



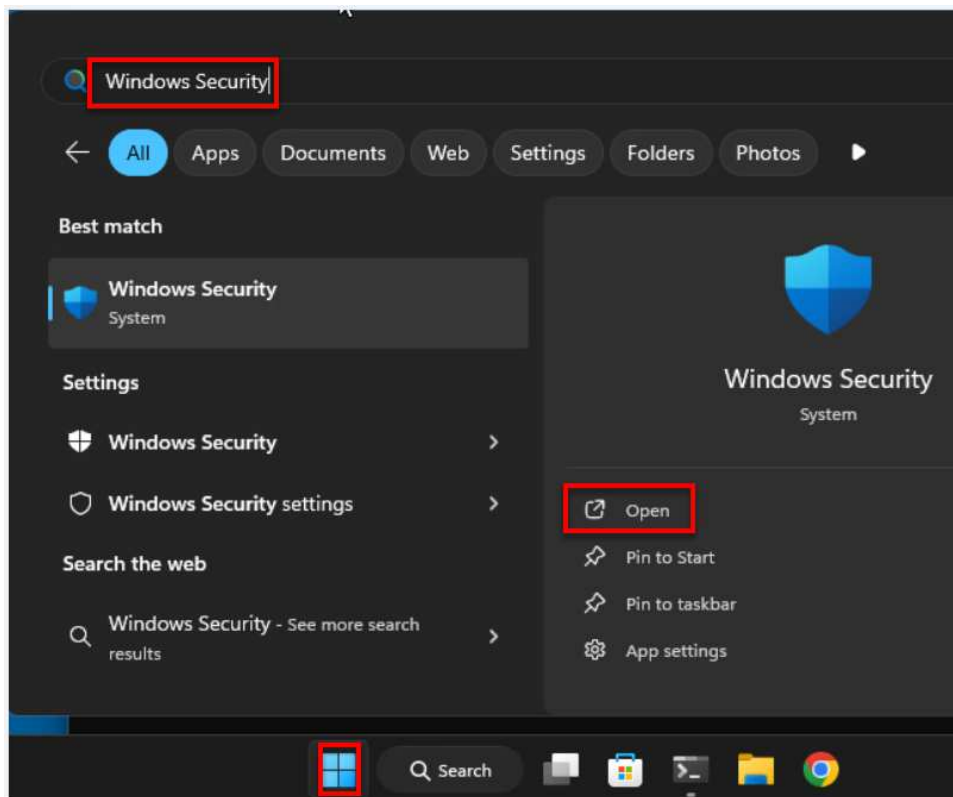
10. In the center pane, scroll to the *Inbound* Rules and right click **Core Networking Diagnostics – ICMP Echo Request (ICMPv4-In)** for the *Private* Network. Select **Enable Rule**. This will allow an incoming ping echo request for the *Win10-OS* system.



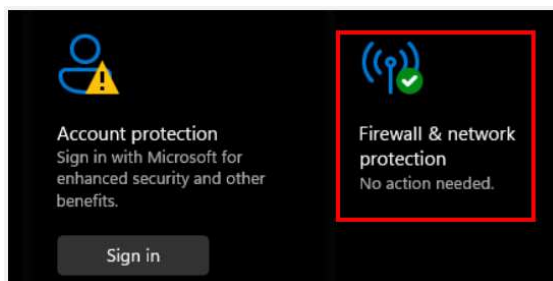
11. Change focus back to the **Win11-OS** system.



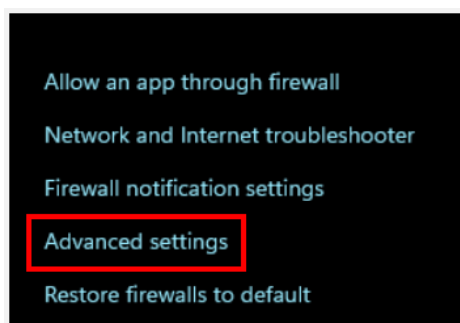
12. Click the **Start** button, in the search for apps textbox, type **Windows Security**, and select **Open**.



13. Select **Firewall & network protection**.



14. Select **Advanced settings**.



15. Select **Yes** to allow changes via *User Account Control*.



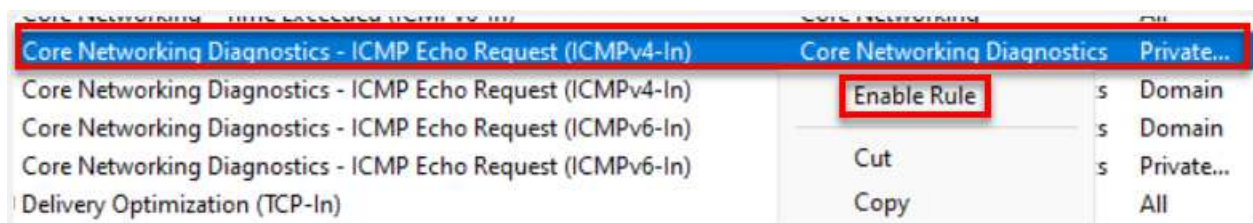
16. Switch to the open *Windows Defender Firewall with Advanced Security* window.



17. Select **Inbound Rules**.



18. In the center pane, scroll to the Inbound Rules and right-click **Core Networking Diagnostics – ICMP Echo Request (ICMPv4-In)** for the *Private Network*. Select **Enable Rule**. This will allow incoming ping echo requests for the Win11-OS system.



19. Change focus to the open *Command Prompt*. Ping the *Win10-OS* system by typing `ping 192.168.1.10`. The echo ping request should process successfully.

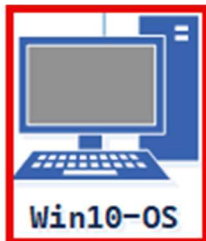
```
ping 192.168.1.10
```

```
C:\Users\sysadmin>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

20. Change focus back to the **Win10-OS** system.



21. In the open *Command Prompt*, ping the *Win11-OS* system by typing `ping 192.168.1.11`. The echo ping request should process successfully.

```
ping 192.168.1.11
```

```
C:\Users\sysadmin>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\sysadmin>
```

22. Leave the *Win10-OS* window open for the next task.

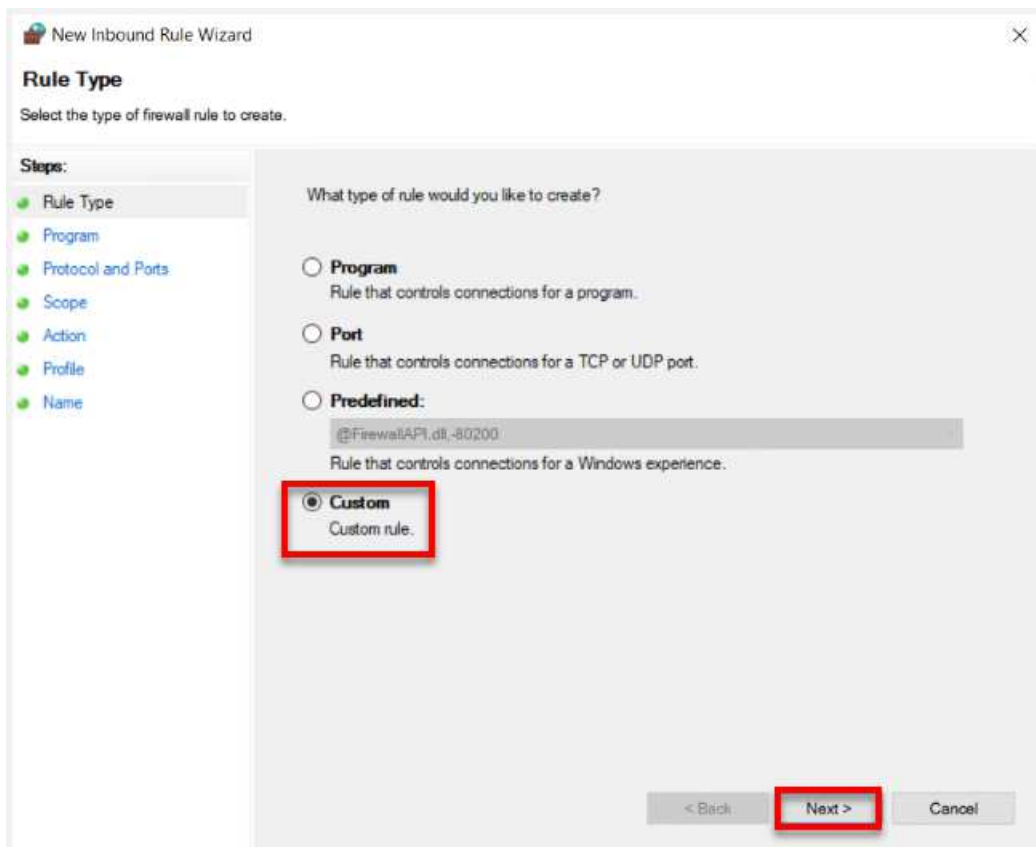
1.8 Creating New Rules with Windows Firewall in Win10-OS

In the previous exercise, you learned how to configure ping as a firewall rule and enable it using existing *ICMP* rules. In this exercise, you will create a custom inbound rule to not allow the *TCP/IP* ping *Echo Request* command. The default action is to enable all *ICMP* traffic, but you will create a custom rule for the *Echo Request*. You will do this for *IPv4*. You will do this on both *Win10-OS* and *Win11-OS* systems.

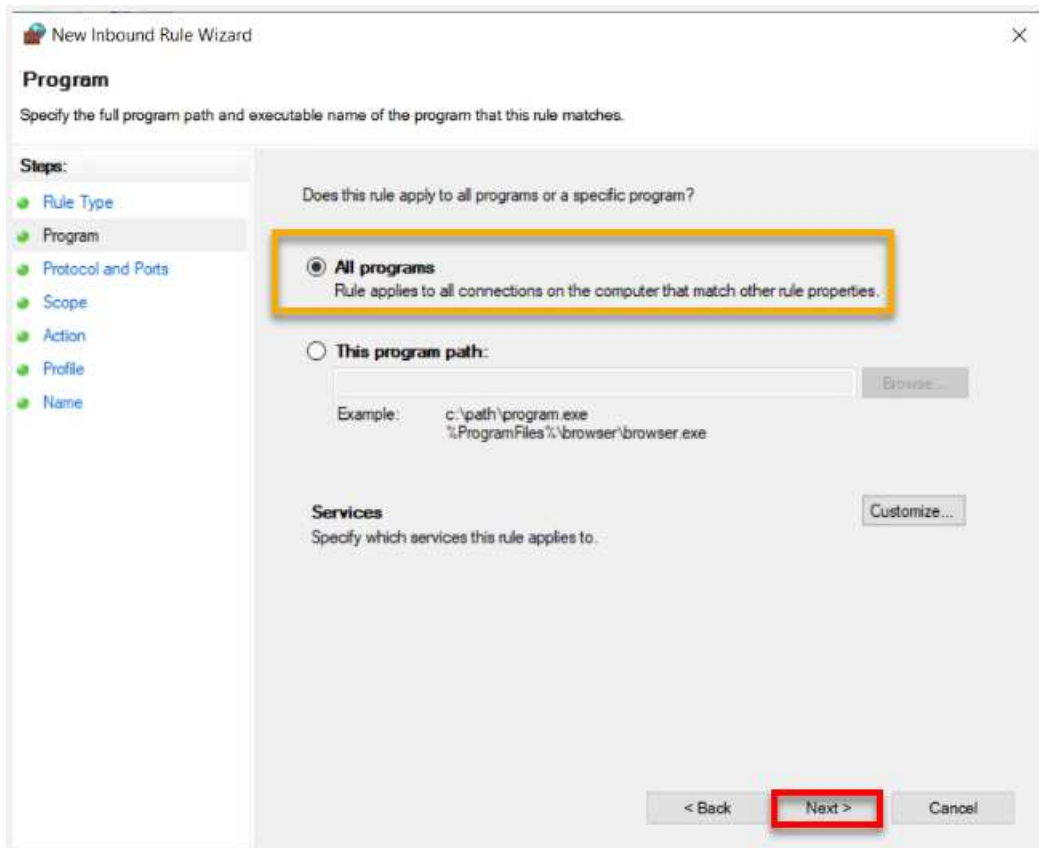
1. While on the *Win10-OS* system, focus on the **Windows Defender Firewall with Advanced Security** window.
2. Right-click on **Inbound Rules** and click **New Rule**.



3. In the *New Inbound Rule Wizard*, select the radio button for **Custom**. Click **Next**.

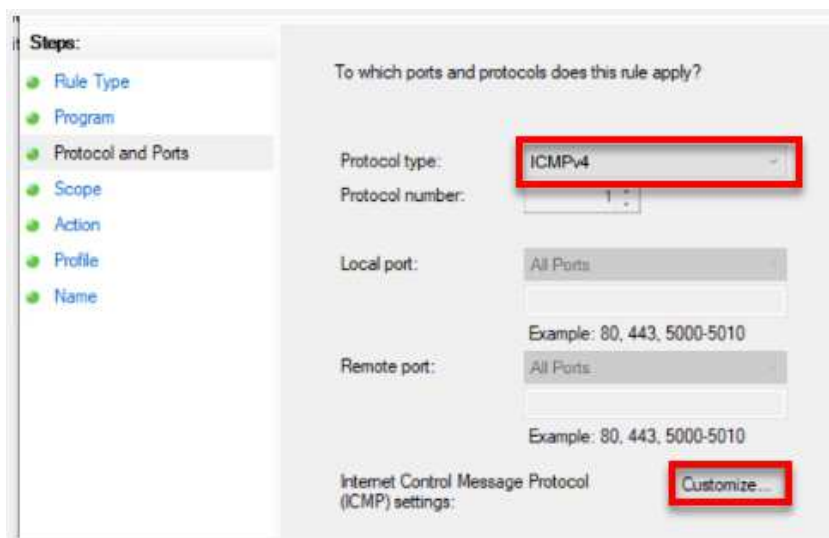


4. On the *Program* step, ensure that the radio button for **All programs** is selected and click **Next**.



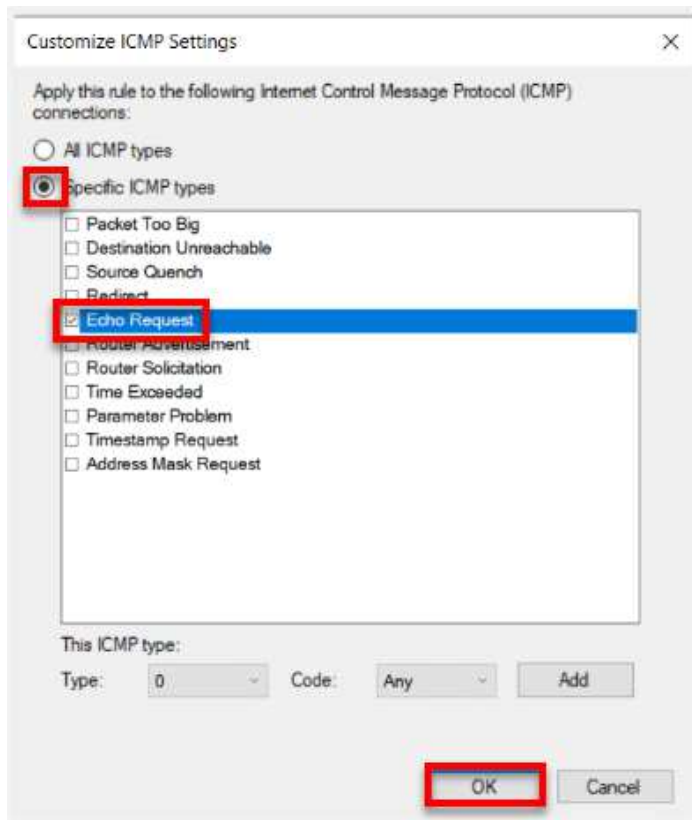
The screenshot shows the 'New Inbound Rule Wizard' window, specifically the 'Program' step. The title bar reads 'New Inbound Rule Wizard'. The main heading is 'Program', with the instruction 'Specify the full program path and executable name of the program that this rule matches.' On the left, a 'Steps' pane lists: Rule Type, Program, Protocol and Ports, Scope, Action, Profile, and Name. The 'Program' step is currently selected. The main area asks 'Does this rule apply to all programs or a specific program?'. There are two radio buttons: 'All programs' (selected and highlighted with a yellow box) and 'This program path:'. The 'All programs' option has the description 'Rule applies to all connections on the computer that match other rule properties.' Below the radio buttons is a text field for 'This program path:' with a 'Browse...' button. An example path is shown: 'c:\path\program.exe' and '%ProgramFiles%\browser\browser.exe'. Below that is a 'Services' section with a 'Customize...' button. At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a red box), and 'Cancel'.

5. On the *Protocol and Ports* step, select **ICMPv4** from the *Protocol type* dropdown menu. Click the **Customize** button.

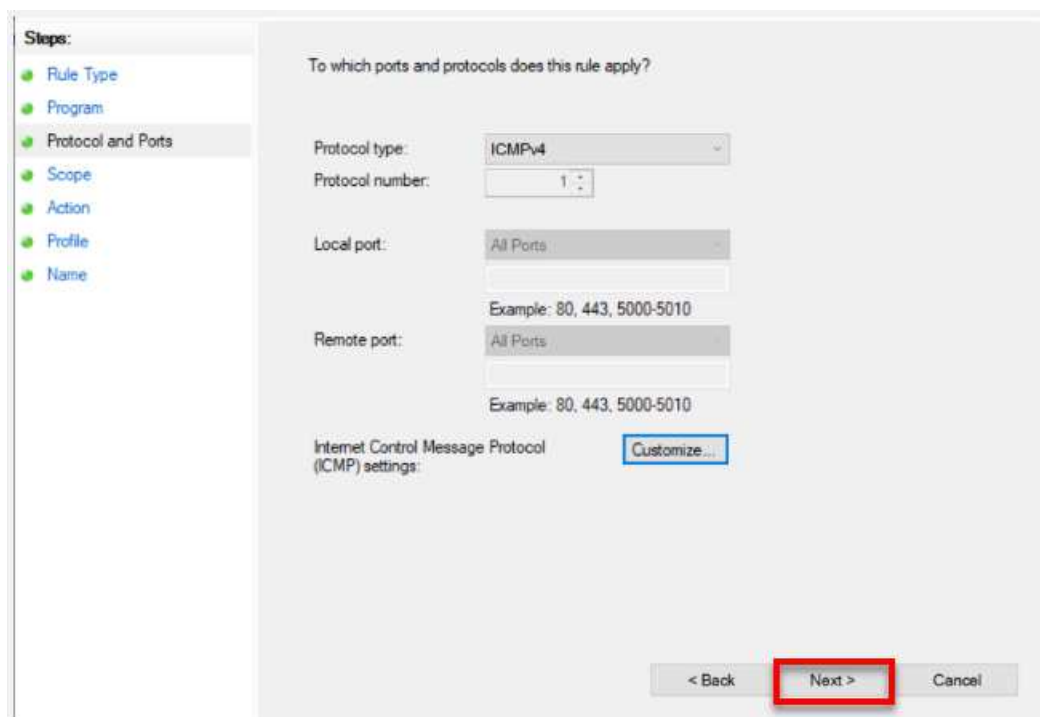


The screenshot shows the 'New Inbound Rule Wizard' window, specifically the 'Protocol and Ports' step. The 'Steps' pane on the left shows 'Protocol and Ports' as the selected step. The main heading is 'To which ports and protocols does this rule apply?'. There are four fields: 'Protocol type:' (a dropdown menu with 'ICMPv4' selected and highlighted with a red box), 'Protocol number:' (a text field with '1' entered), 'Local port:' (a dropdown menu with 'All Ports' selected), and 'Remote port:' (a dropdown menu with 'All Ports' selected). Below these fields are two example text boxes: 'Example: 80, 443, 5000-5010'. At the bottom, there is a section for 'Internet Control Message Protocol (ICMP) settings:' and a 'Customize...' button (highlighted with a red box).

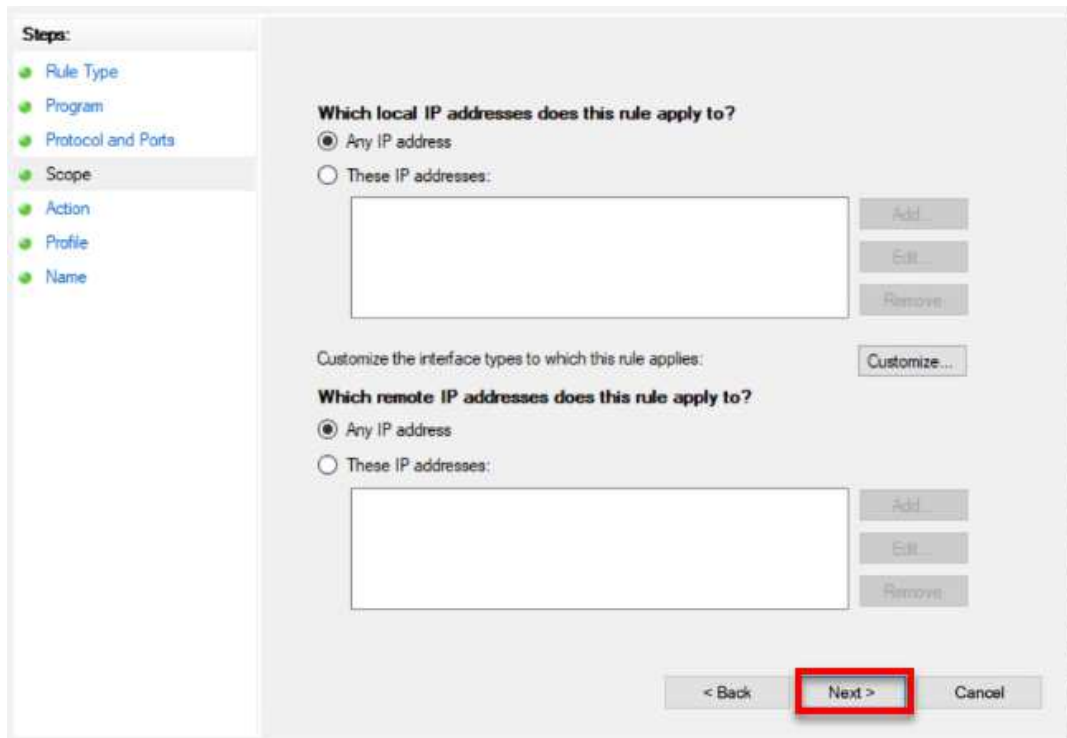
6. In the *Customize ICMP Settings* window, select the radio button for **Specific ICMP types**. From the list of *ICMP* types, check the checkbox next to **Echo Request** to enable it. Click **OK**.



7. Back on the *New Inbound Rule Wizard*, click **Next**.

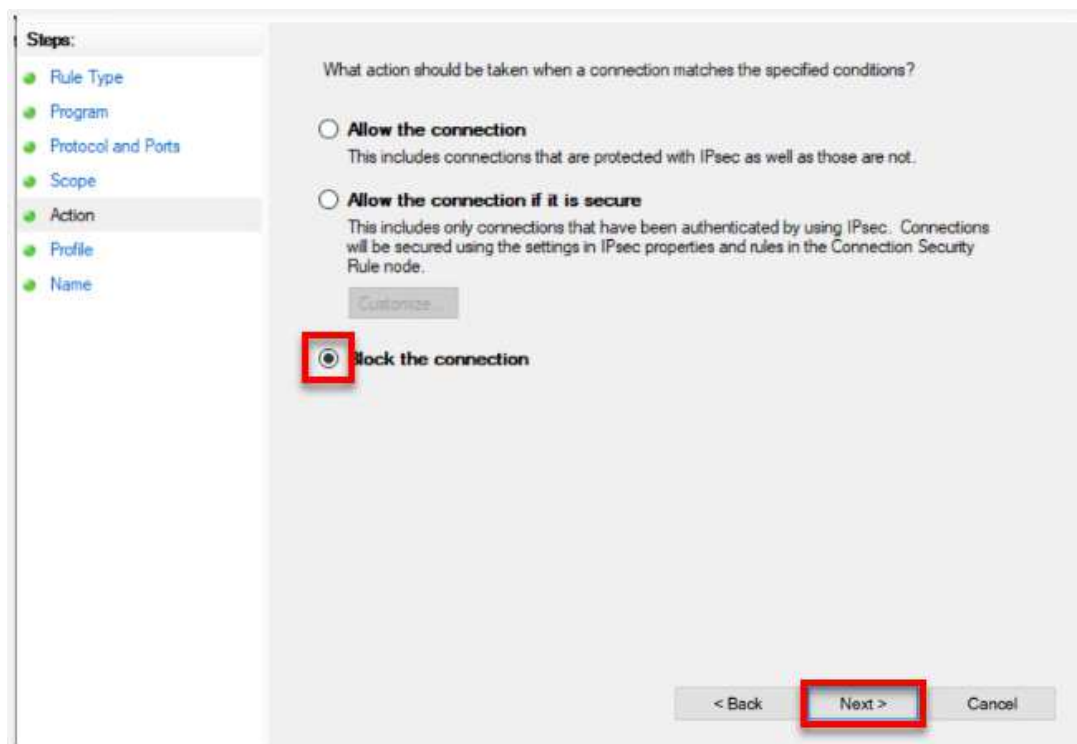


8. On the *Scope* step, you can configure specific IP addresses to which the system will respond to. Leave the defaults and click **Next**.



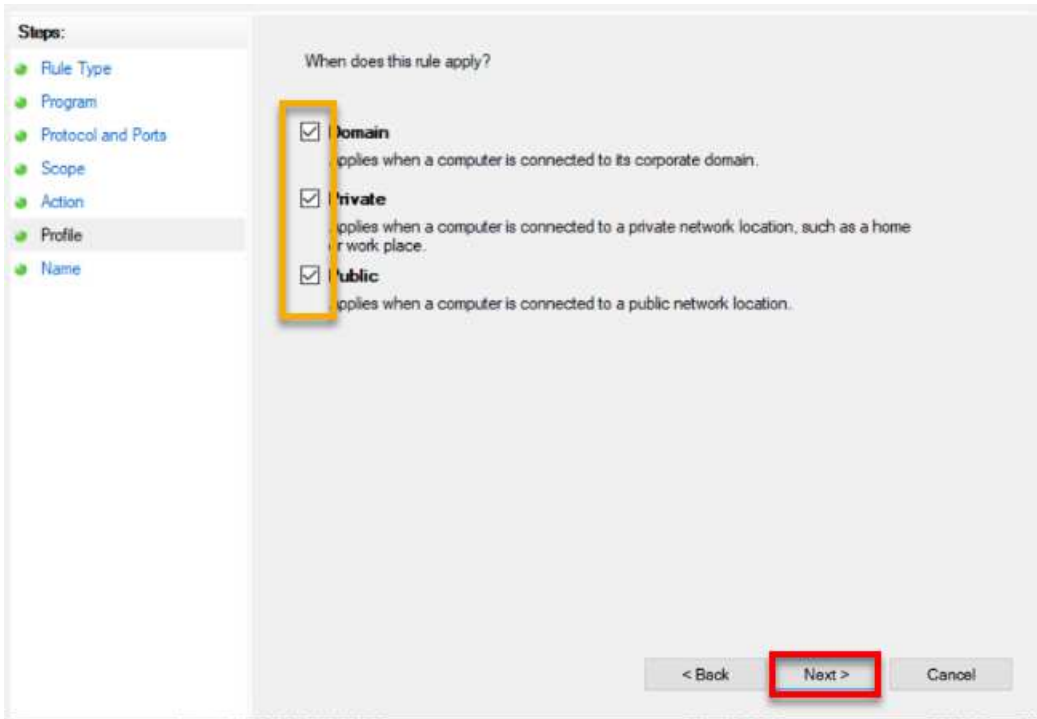
The screenshot shows the 'Scope' step of a configuration wizard. On the left, a 'Steps' sidebar lists: Rule Type, Program, Protocol and Ports, Scope (highlighted), Action, Profile, and Name. The main area is titled 'Which local IP addresses does this rule apply to?'. It has two radio buttons: 'Any IP address' (selected) and 'These IP addresses:'. Below the second option is a text box and buttons for 'Add...', 'Edit...', and 'Remove...'. Below this is a section 'Customize the interface types to which this rule applies:' with a 'Customize...' button. Another section 'Which remote IP addresses does this rule apply to?' has the same radio buttons and controls. At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a red box), and 'Cancel'.

9. On the *Action* step, ensure that **Block the connection** is selected. Click **Next**.

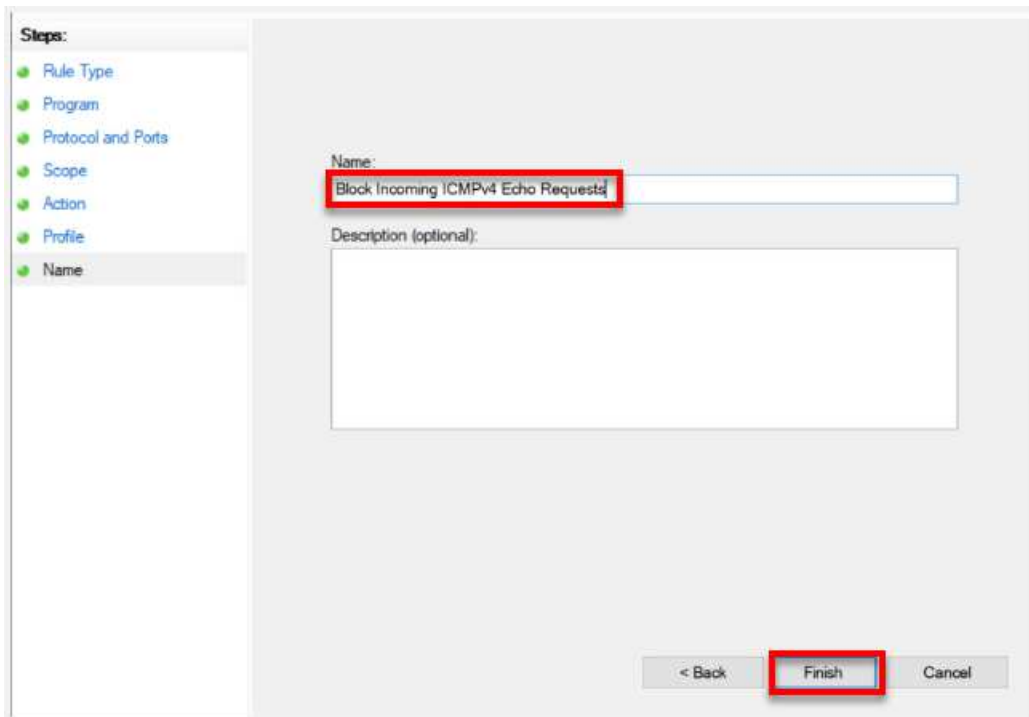


The screenshot shows the 'Action' step of the configuration wizard. The 'Steps' sidebar on the left is the same as in the previous image, with 'Action' now highlighted. The main area is titled 'What action should be taken when a connection matches the specified conditions?'. It has two radio buttons: 'Allow the connection' (with a sub-note: 'This includes connections that are protected with IPsec as well as those are not.') and 'Allow the connection if it is secure' (with a sub-note: 'This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.'). Below these is a 'Customize...' button. The third option, 'Block the connection', is selected and highlighted with a red box. At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a red box), and 'Cancel'.

10. On the *Profile* step, this page allows you to control when the rule is active in which network types. Leave the defaults and click **Next**.



11. On the *Name* step, type **Block Incoming ICMPv4 Echo Requests** in the *Name* text field. Click **Finish**.



12. Change focus to the **Win11-OS** system.



13. At the command prompt window of the *Win11-OS* system, enter the command below. Can the *Win11-OS* system still ping successfully?

```
ping 192.168.1.10
```

```
C:\Users\sysadmin>ping 192.168.1.10

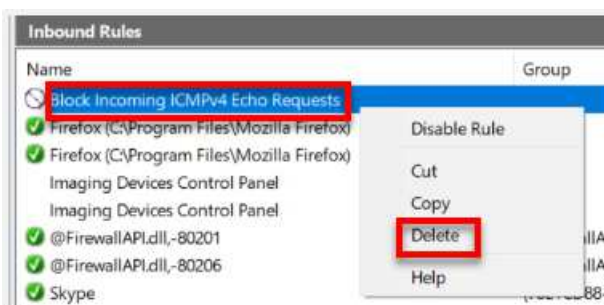
Pinging 192.168.1.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

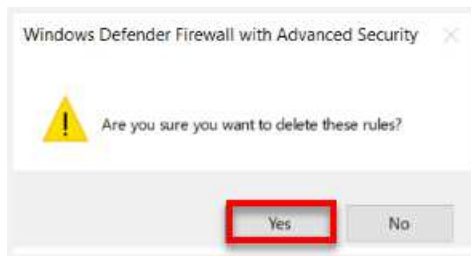
14. Change focus to the **Win10-OS** system.



15. Go back to the **Windows Defender Firewall with Advanced Security** window on the *Win10-OS* system, right-click on the **Block Incoming ICMPv4 Echo Requests** rule, and select **Delete**.

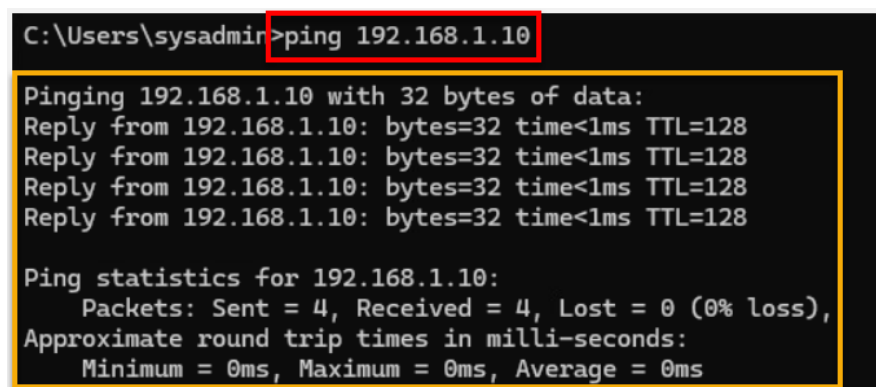


16. Select **Yes** to delete this rule.



17. Change focus back to the open *Windows Defender Firewall with Advanced Security* window. Right-click the *Inbound Rule* **Core Networking Diagnostics – ICMP Echo Request (ICMPv4-In)** for the *Private Network*, and notice that this rule wasn't disabled by default.
18. Change focus to the **Win11-OS** system.
19. At the command prompt window of the *Win11-OS* system, enter the command below. Notice that the Win11-OS system can now ping successfully.

```
ping 192.168.1.10
```



```
C:\Users\sysadmin>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

20. Close the **command prompt**, **Windows Defender Firewall with Advanced Security**, and **Windows Security** windows.
21. Leave the *Win11-OS* window open to continue with the next task.

2 Windows Encryption

Windows 10 and 11 offer two primary encryption methods:

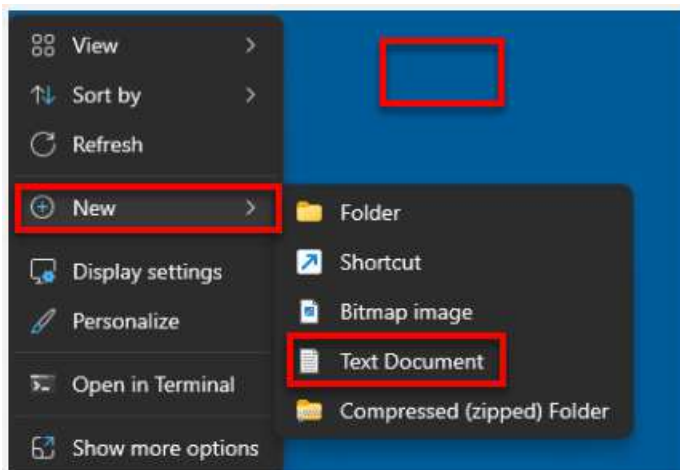
1. **Encrypting File System (EFS):** This feature encrypts individual files and folders on a drive. It's available in both Home and Pro/Enterprise editions.
2. **BitLocker Drive Encryption:** This robust solution encrypts the entire drive, providing a higher level of security. It's primarily available in Pro and Enterprise editions.

Encryption is a process that transforms data into an unreadable format, protecting it from unauthorized access.

2.1 Encrypting File System in Win11-OS

In this exercise, you will use *EFS* to encrypt a file you create on the *Win11-OS* system. *EFS-encrypted* files can only be accessed by the user account that encrypted them. The encryption is transparent. If the user account that encrypted the files is logged in, they can access the files without any additional authentication. If another user account is logged in, the files will not be accessible.

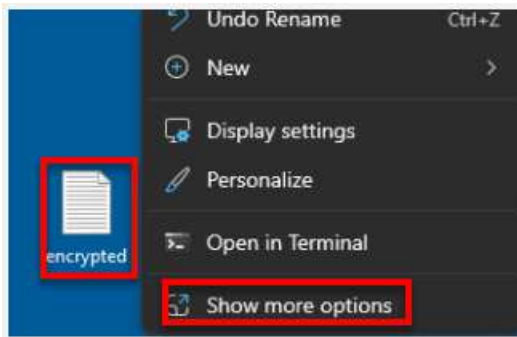
1. Right-click on the *Win11-OS* desktop and select **New > Text Document**.



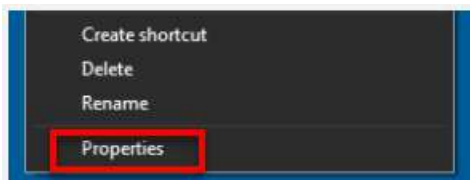
2. Name the new file **encrypted**.



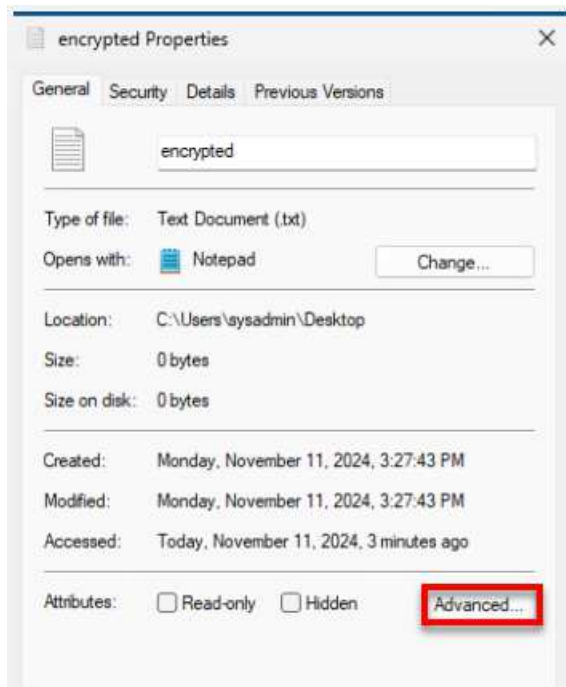
3. Right-click the **encrypted** text file and click on **Show more options**.



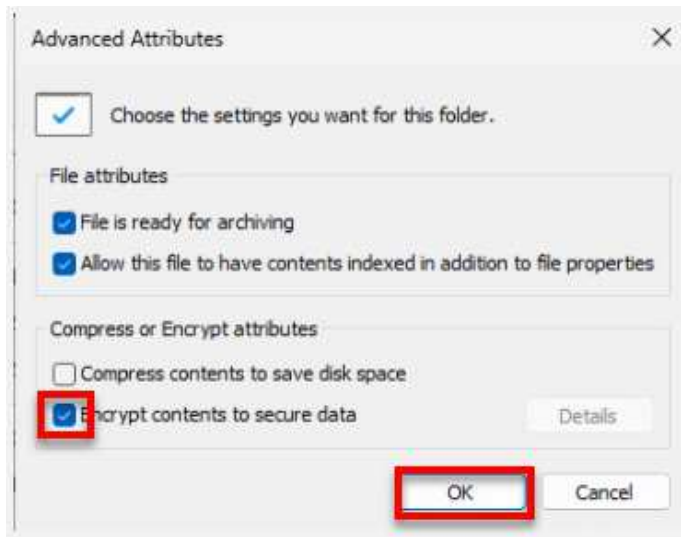
4. Select **Properties**.



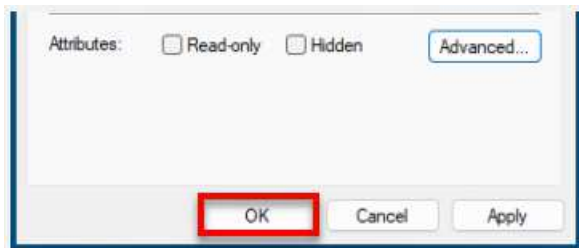
5. On the *General* tab, click on the **Advanced** button.



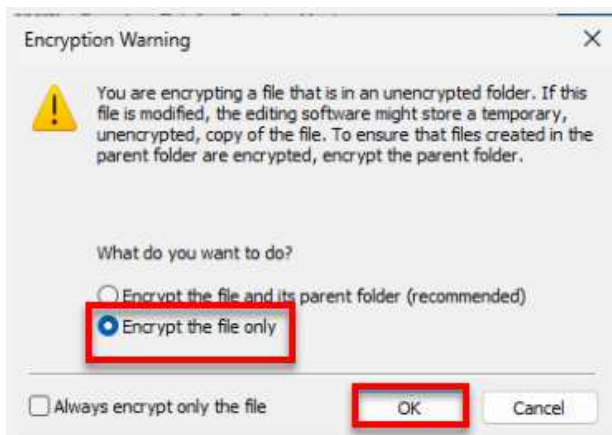
6. On the *Advanced Attributes* window, check the checkbox for **Encrypt contents to secure data**. Click **OK**.



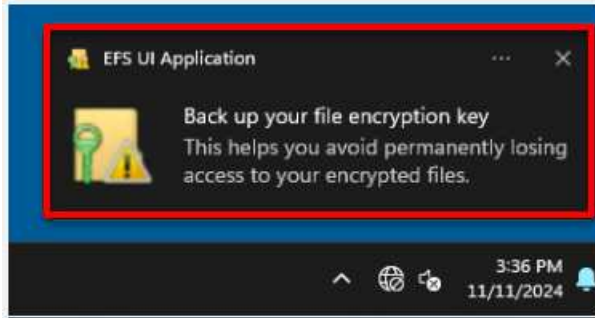
7. Back on the *encrypted Properties* window, click **OK**.



8. A pop-up will appear. This is where you can choose how extensive the encryption is for the data. Select the radio button next to **Encrypt the file only**. Click **OK**.



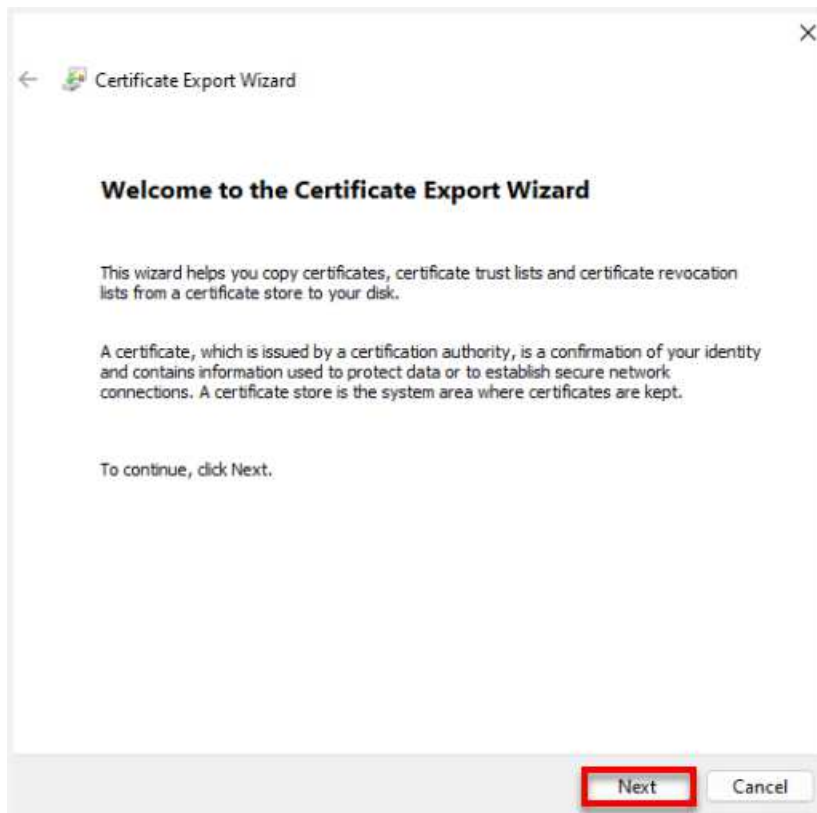
9. A notification appears. Click on the **Back up your encryption key** notification icon to back up the key.



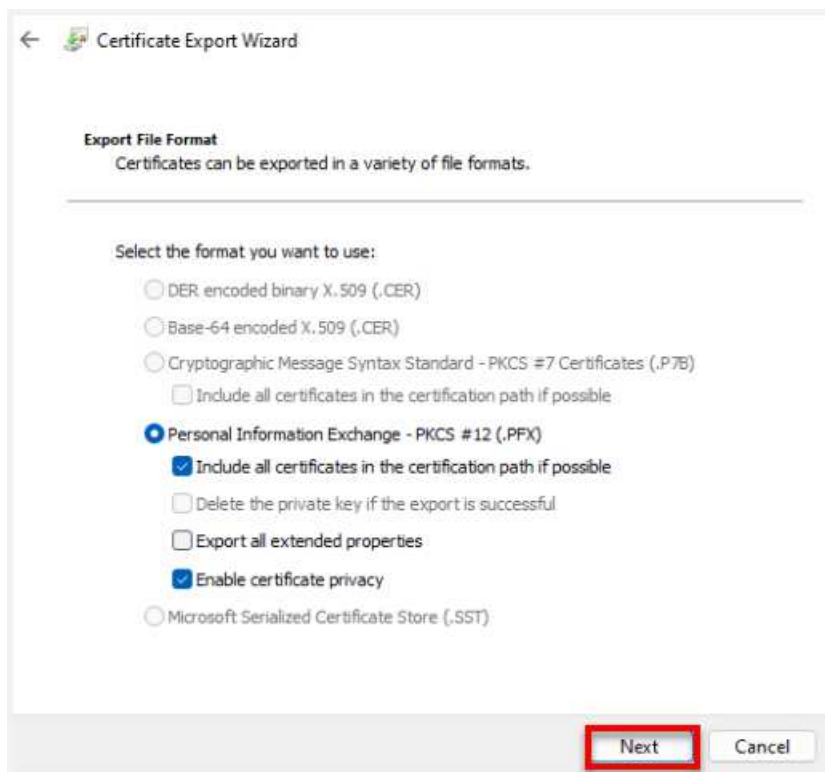
10. In the *Encrypting File System* window, click **Back up now (recommended)**.



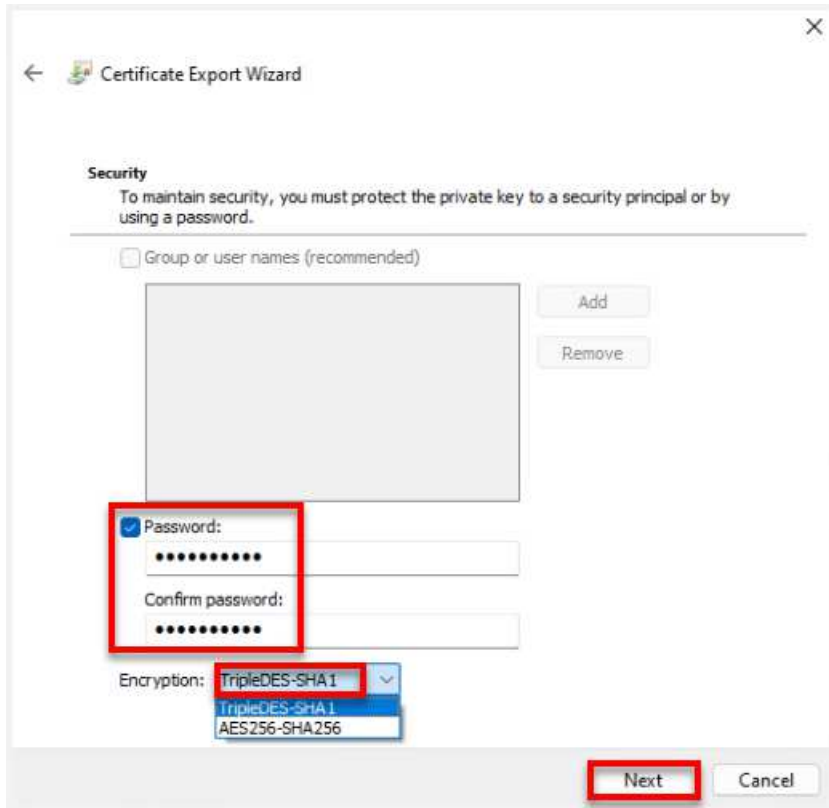
11. In the *Certificate Export Wizard*, click **Next**.



12. Read the selections and then accept the default selection for the file format for the exported encryption certificate and key. Click **Next**.



13. Enable the Password, and type `mypassword` in the *Password* field. Type it again in the confirmation field. Select **TripleDES-SHA1** as *Encryption Type*. Click **Next**.



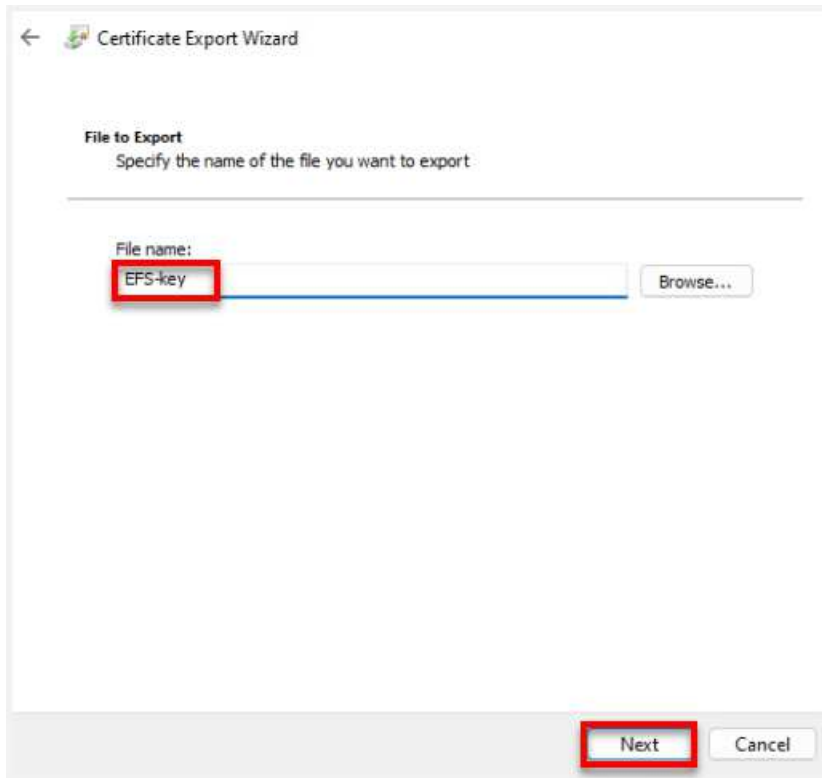
The image shows the 'Certificate Export Wizard' window, specifically the 'Security' step. The window title is 'Certificate Export Wizard'. Below the title bar, there is a back arrow and the text 'Certificate Export Wizard'. The main heading is 'Security'. Below this, a message states: 'To maintain security, you must protect the private key to a security principal or by using a password.' There are two options: 'Group or user names (recommended)' which is unchecked, and 'Password:' which is checked. The 'Password:' option is highlighted with a red box. Below the 'Password:' checkbox, there are two password input fields: 'Password:' and 'Confirm password:'. Both fields contain a series of dots representing masked text. Below the password fields, there is an 'Encryption:' dropdown menu. The dropdown is open, showing three options: 'TripleDES-SHA1' (selected), 'TripleDES-SHA1', and 'AES256-SHA256'. The 'TripleDES-SHA1' option is highlighted with a red box. At the bottom right of the window, there are two buttons: 'Next' and 'Cancel'. The 'Next' button is highlighted with a red box.



When exporting a certificate with its private key, the Certificate Export Wizard typically offers two encryption types: AES 256-bit and Triple DES. AES 256-bit is the more secure option, providing robust protection for sensitive private keys. However, if compatibility with specific software or systems is a concern, Triple DES can be used, although it's less secure.

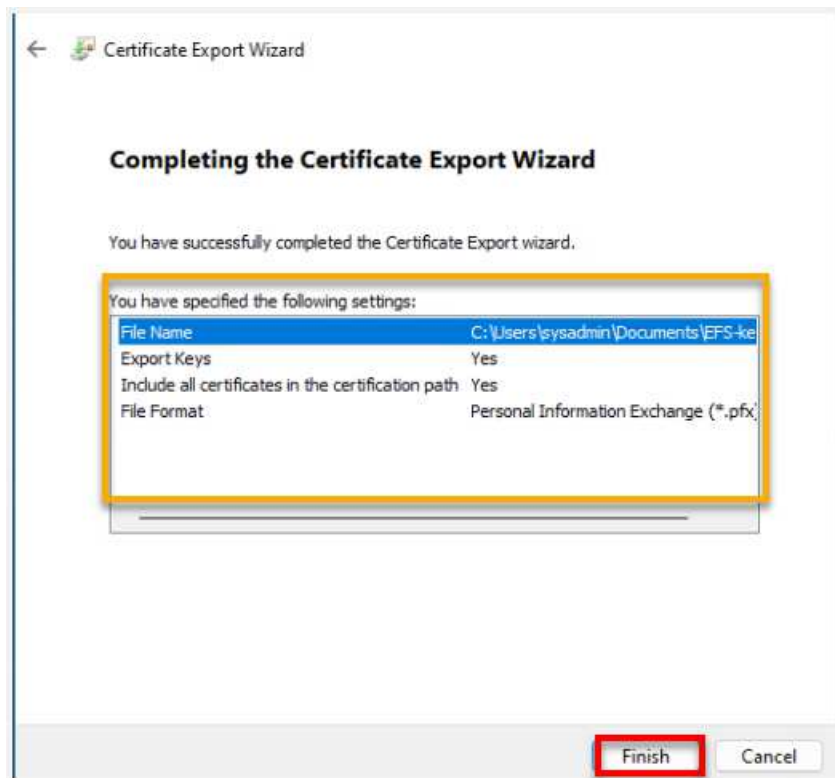
To ensure optimal security, it's crucial to choose a strong password for the exported certificate and store it in a secure location. Additionally, keeping your system and security software up-to-date is essential to address potential vulnerabilities. By following these best practices and selecting the appropriate encryption type, you can safeguard your certificate and its associated private key.

14. Type EFS-key in the *File name* field. Click **Next**.



The screenshot shows the 'Certificate Export Wizard' window. The title bar says 'Certificate Export Wizard'. Below the title bar, there is a section titled 'File to Export' with the instruction 'Specify the name of the file you want to export:'. Below this, there is a text box labeled 'File name:' containing the text 'EFS-key'. To the right of the text box is a 'Browse...' button. At the bottom right of the window, there are two buttons: 'Next' and 'Cancel'. The 'Next' button is highlighted with a red rectangle.

15. Review the configurations and take note of the *File Name* location. Click **Finish**.



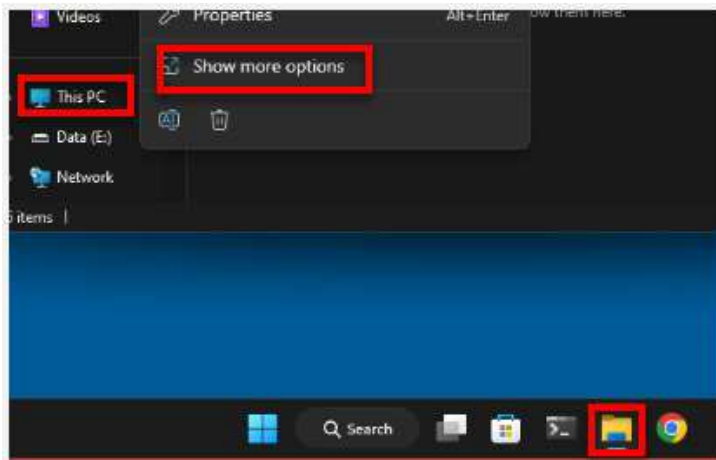
The screenshot shows the 'Completing the Certificate Export Wizard' screen. The title bar says 'Certificate Export Wizard'. Below the title bar, there is a section titled 'Completing the Certificate Export Wizard' with the message 'You have successfully completed the Certificate Export wizard.' Below this, there is a box titled 'You have specified the following settings:' containing a table of settings. The table has two columns: the setting name and the value. The settings are: 'File Name' (C:\Users\sysadmin\Documents\EFS-key), 'Export Keys' (Yes), 'Include all certificates in the certification path' (Yes), and 'File Format' (Personal Information Exchange (*.pfx)). The 'File Name' row is highlighted with a blue background. Below the table, there is a 'Finish' button and a 'Cancel' button. The 'Finish' button is highlighted with a red rectangle.

You have specified the following settings:	
File Name	C:\Users\sysadmin\Documents\EFS-key
Export Keys	Yes
Include all certificates in the certification path	Yes
File Format	Personal Information Exchange (*.pfx)

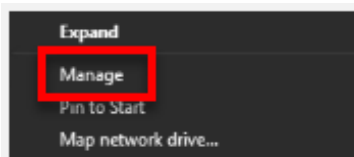
16. The export is successful; click **OK**.



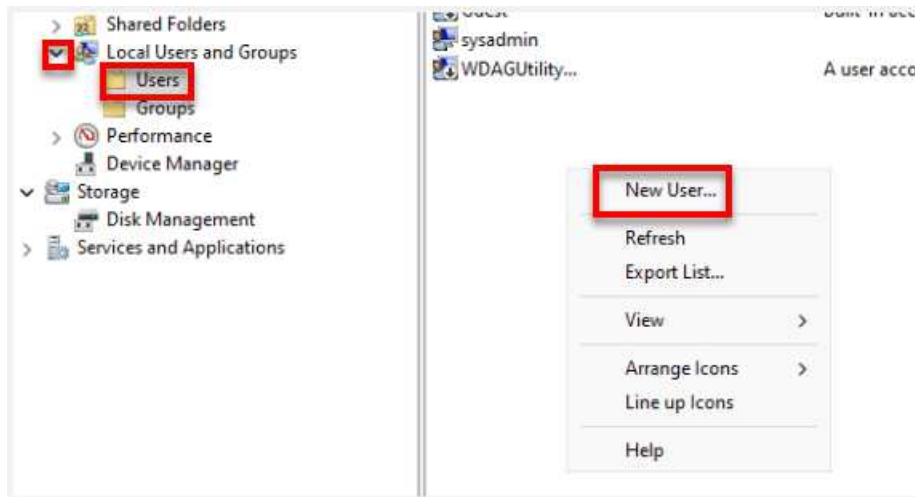
17. Using *EFS*, you have just encrypted a single file on a per-user basis for the *sysadmin* account. Encrypted files can only be accessed by the user who created them. The encryption is transparent. Since you are logged in as *a sysadmin*, *you can* access the file without any additional authentication. If another user account is logged in, the file will not be accessible. Double-click on the **encrypted** file, and notice that it opens.
18. Close the **Notepad** window.
19. Select **File Explorer** on the taskbar. Right-click **This PC** and select **Show more options**.



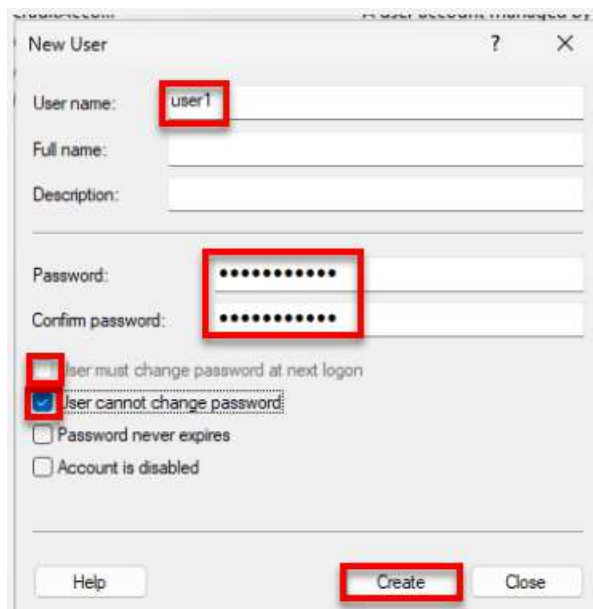
20. Select **Manage**.



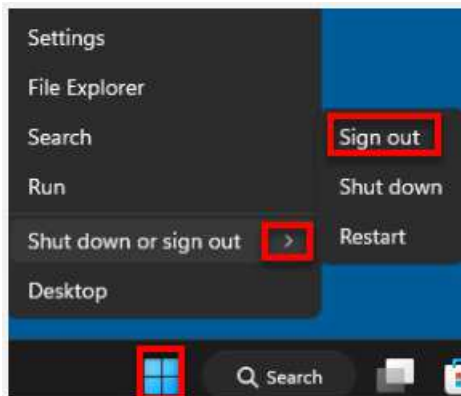
21. Select the **Arrow** next to *Local Users and Groups*. Select **Users**. Right-click the empty space in the middle pane and select **New User**.



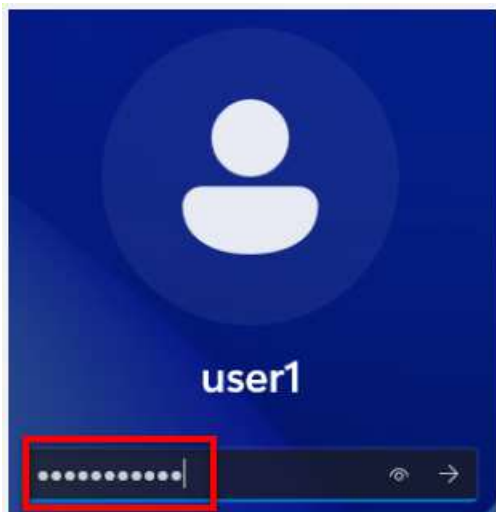
22. Enter the following information. User name **user1**. Password **SuperSecret**. Uncheck **User must change password at next logon** and select **User cannot change password**. Select **Create**.



23. Close all open windows. Right-click the Start button, click the arrow next to Shut down or sign out, and click **Sign out**.



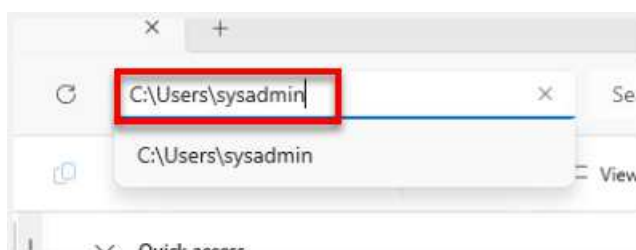
24. Log in as **user1** with SuperSecret as the password. After *user1* is logged in, select **Next** and **Accept** to set up the user1 account.



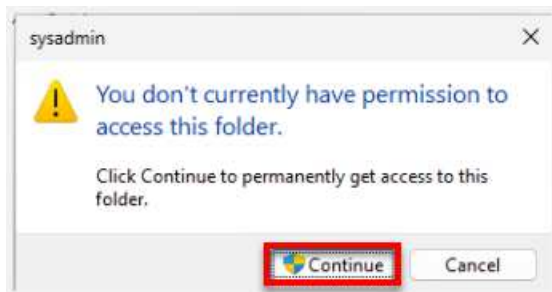
25. Click on the **Windows Explorer** icon located in the taskbar.



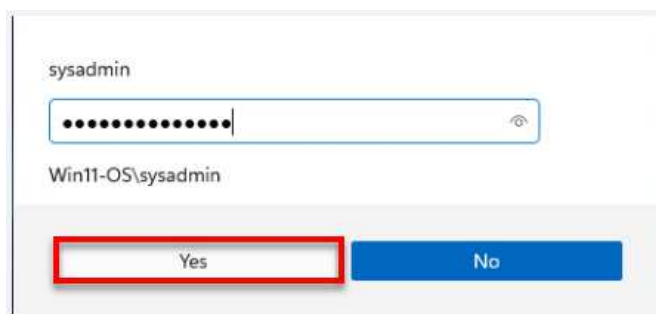
26. Navigate to **C:\Users\sysadmin** and hit **Enter** on your keyboard.



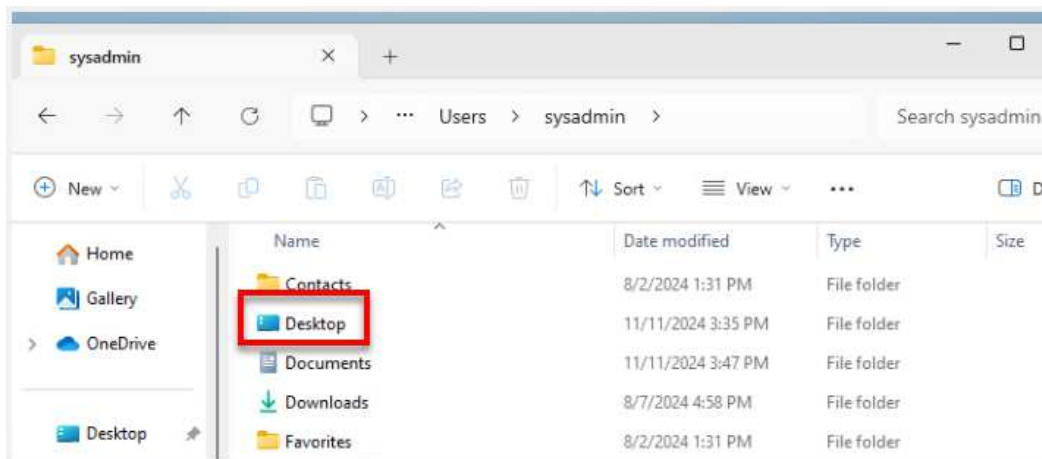
27. A notification window pops up with a notification stating you don't currently have permission to access this folder. Select **Continue**



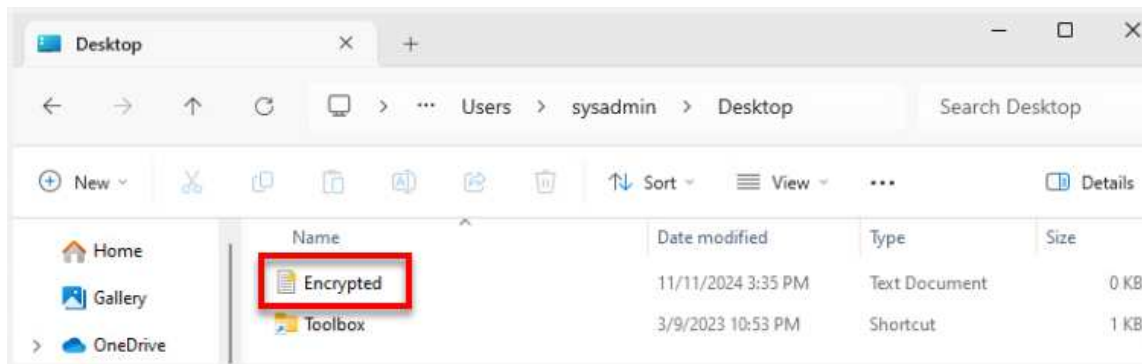
28. Enter the sysadmin password, type **NDGlabpass123!**, and select **Yes**.



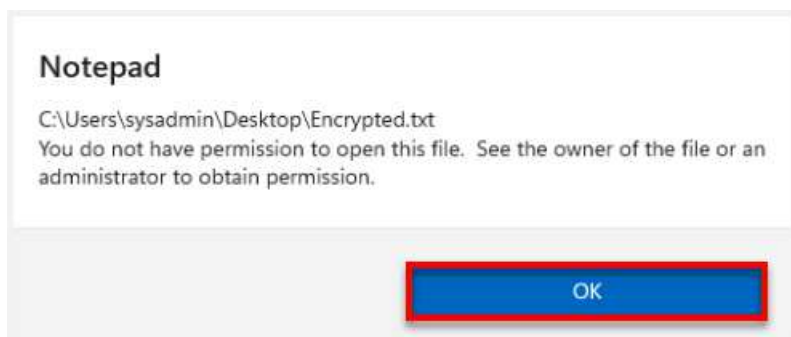
29. Double-click the **Desktop** folder.



30. Double-click the **encrypted** file.



31. Notice that a message appears stating that you do not have permission to open this file. Click **OK**.



32. Log out of **user1**.

2.2 Enabling BitLocker without TPM in Win10-OS

BitLocker is a powerful tool built into Windows that allows you to encrypt an entire hard drive, providing robust security for your data. This is especially crucial for shared systems and mobile devices.

To use BitLocker, your Windows system typically requires a Trusted Platform Module (TPM) chip and at least two partitions on the storage drive. The TPM chip adds an extra layer of security by verifying your device's hardware, software, and firmware before granting access to the encrypted drive.

However, if your system lacks a TPM, there's a workaround available in Windows 10. By enabling additional authentication at startup through the Local Group Policy Editor, you can bypass the TPM requirement and use BitLocker. While this method is functional, it's important to note that full-disk encryption with a TPM is generally considered more secure.

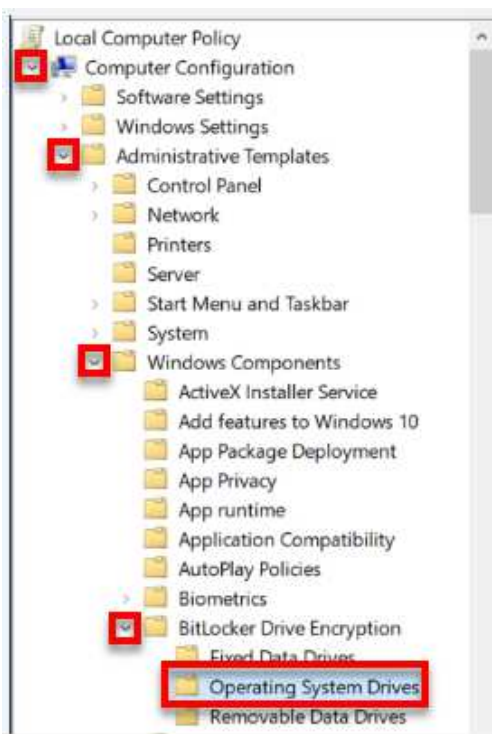
1. Change focus to the **Win10-OS** system.



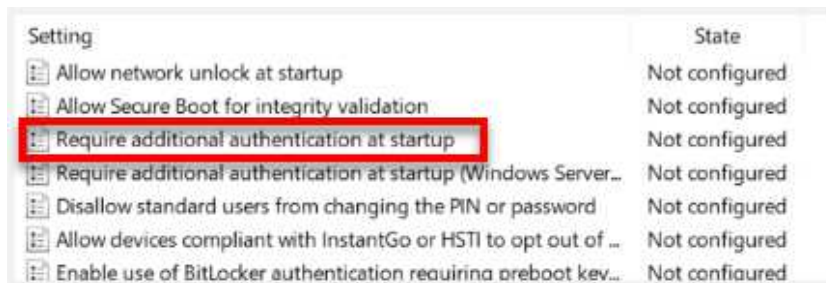
2. If not logged in, log in to the *sysadmin* account with password `NDGlabpass123!`. And click on the **Windows Search** icon on the *Win10-OS* VM. Type **gpedit.msc** in the search field, followed by pressing the **Enter** key.



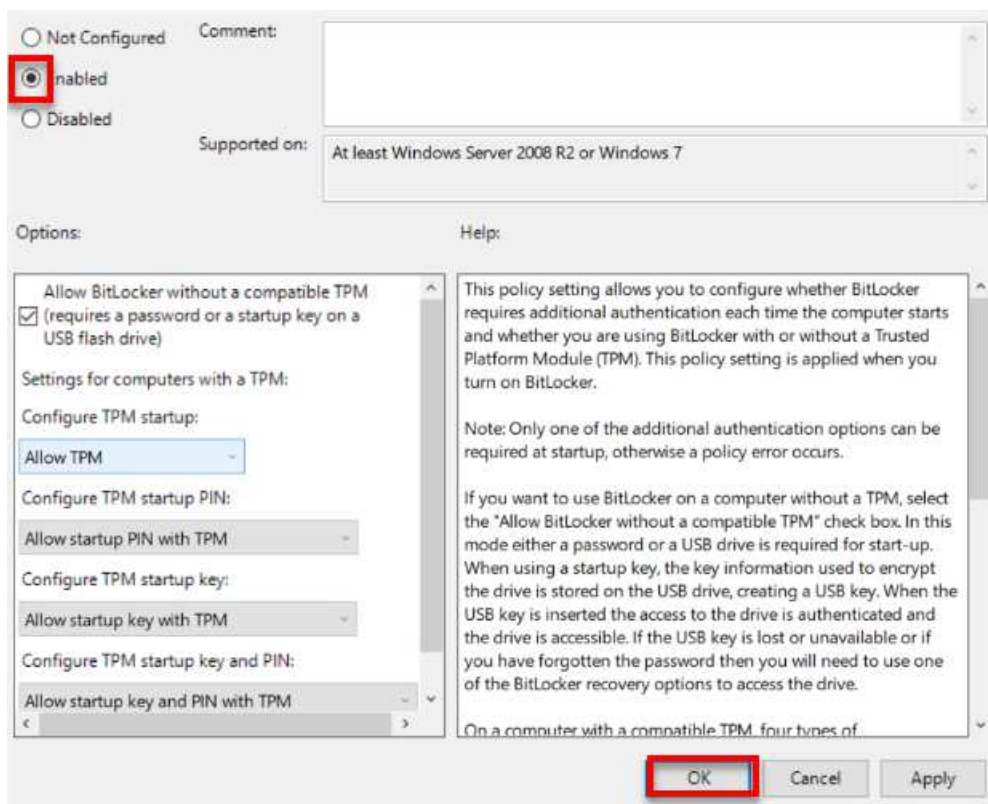
3. In the *Local Group Policy Editor* window, navigate to **Computer Configuration > Administrative Templates > Windows Components > BitLocker Drive Encryption > Operating System Drives**.



4. In the right pane, double-click on **Require additional authentication at startup**.



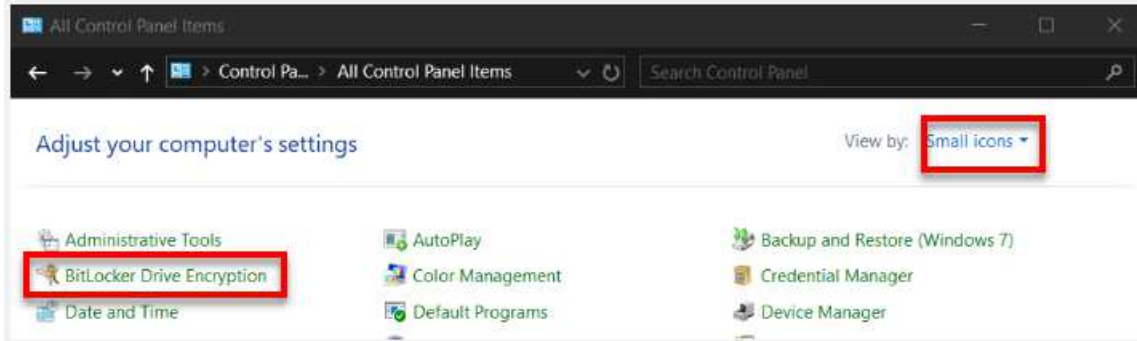
5. In the pop-up window, select the radio button for **Enabled** and click **OK**.



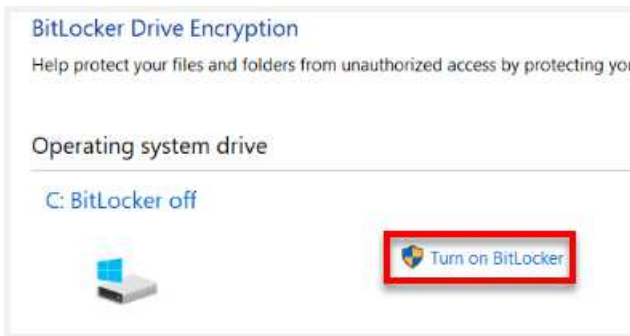
6. Close the **Local Group Policy Editor** window.
7. In the search box of the *Taskbar*, type **Control Panel** and hit **Enter**.



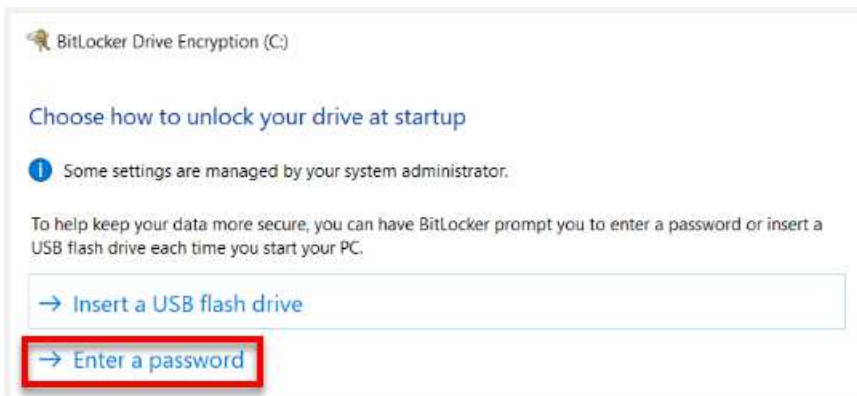
8. In the *Control Panel* window, change the view to **Small icons** and click on **BitLocker Drive Encryption**.



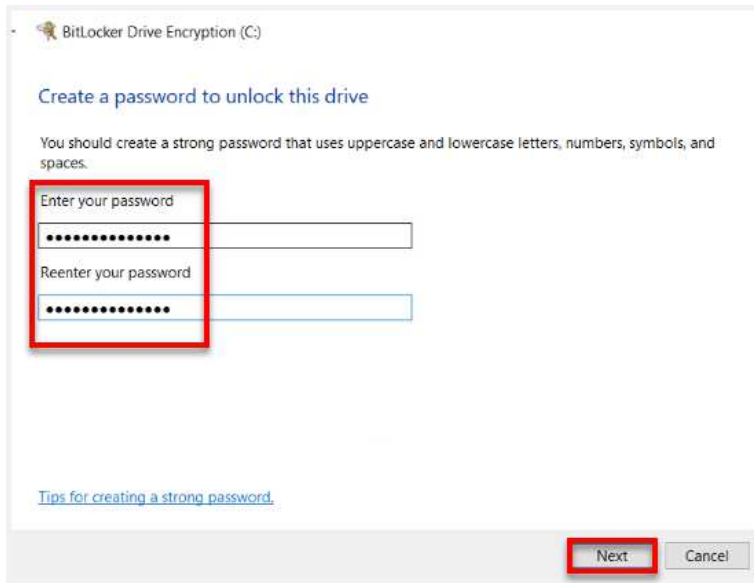
9. Click **Turn on BitLocker**.



10. In the pop-up window, let it scan the system and then click **Enter a password**.



11. Type **NDGlabpass123!** in both password fields. Click **Next**.



BitLocker Drive Encryption (C:)

Create a password to unlock this drive

You should create a strong password that uses uppercase and lowercase letters, numbers, symbols, and spaces.

Enter your password

Reenter your password

[Tips for creating a strong password.](#)

Next Cancel

12. When asked how to back up your recovery key, click **Print the recovery key**.



BitLocker Drive Encryption (C:)

How do you want to back up your recovery key?

Some settings are managed by your system administrator.

A recovery key can be used to access your files and folders if you're having problems unlocking your PC. It's a good idea to have more than one and keep each in a safe place other than your PC.

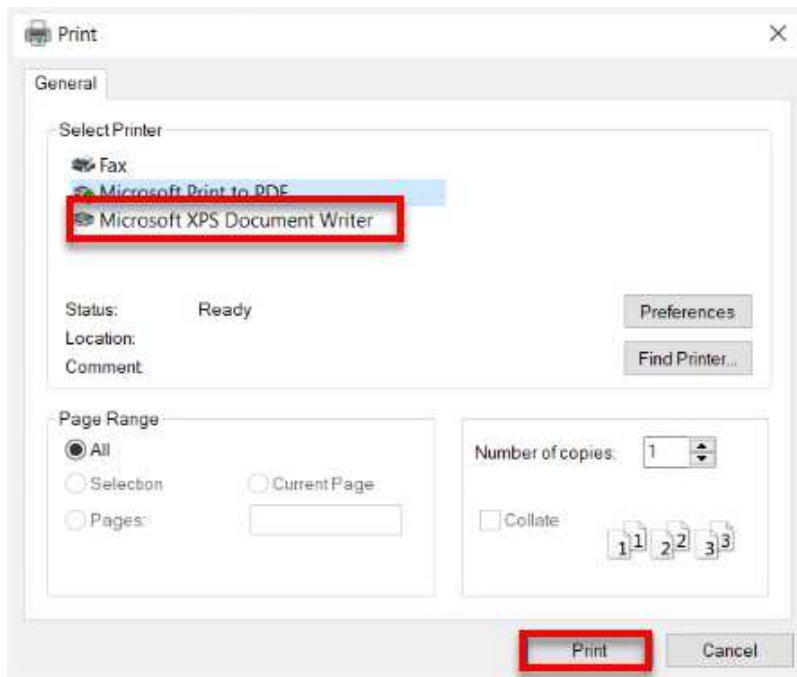
→ Save to your Microsoft account

→ Save to a USB flash drive

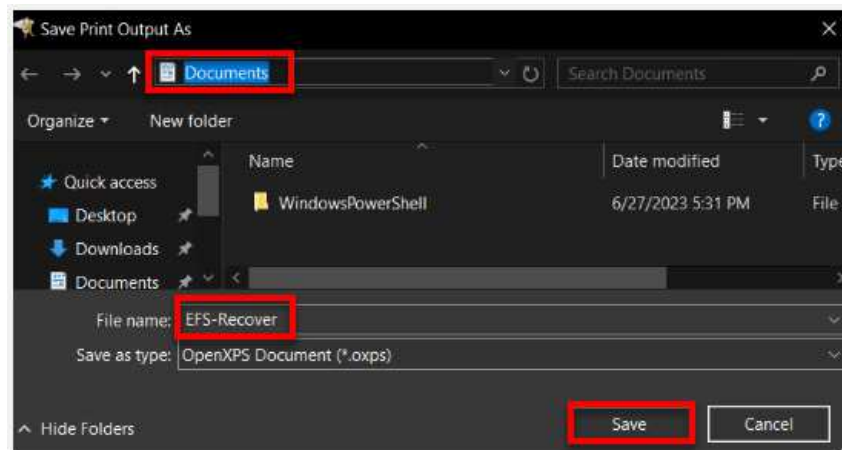
→ Save to a file

→ Print the recovery key

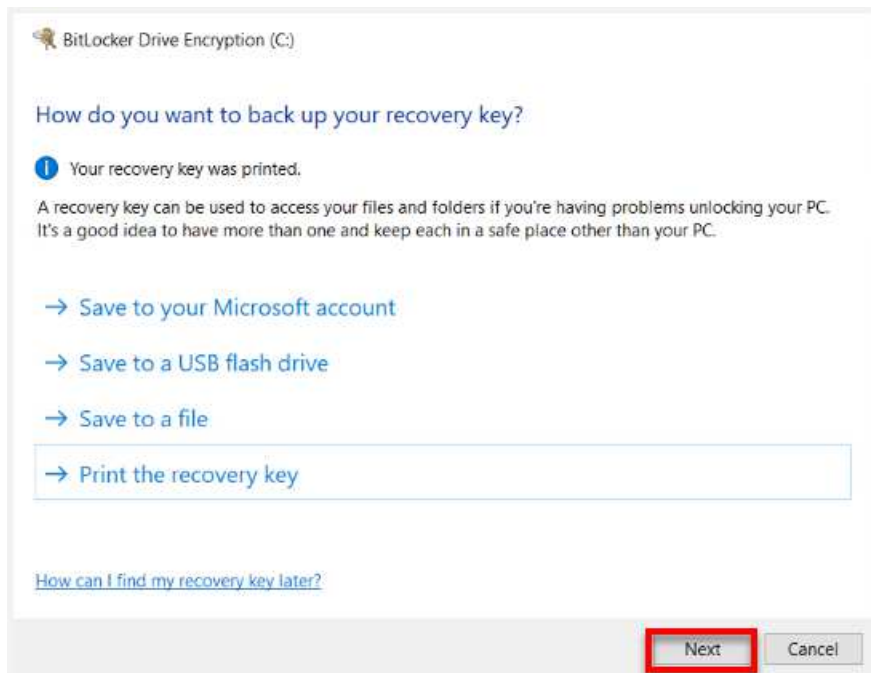
13. Select the **Microsoft XPS Document Writer** print option. Click **Print**.



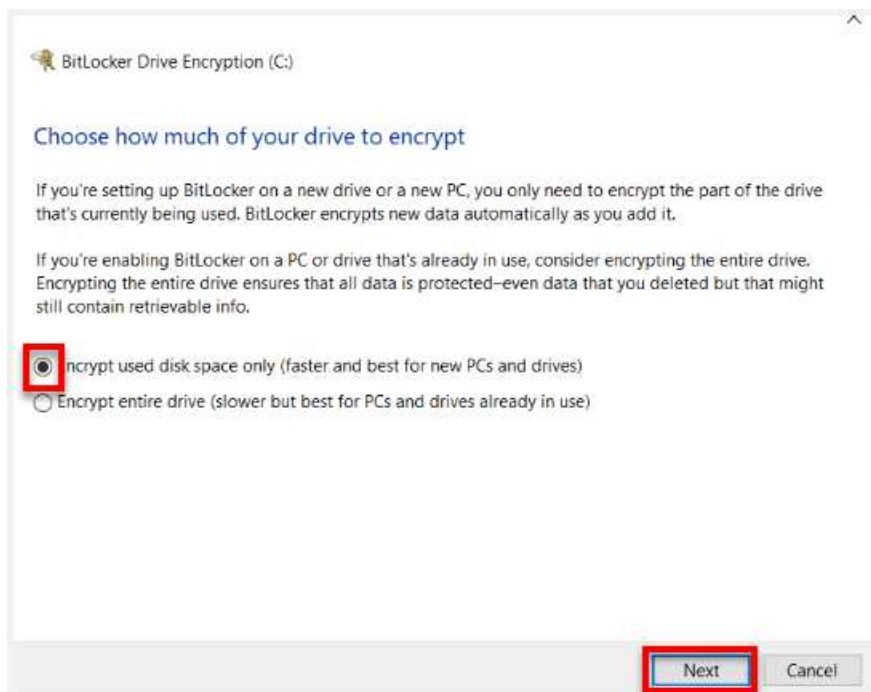
14. In the *Save Print Output As* window, name the file **EFS-Recover** and click **Save** to save the file in the default user's *Documents* location.



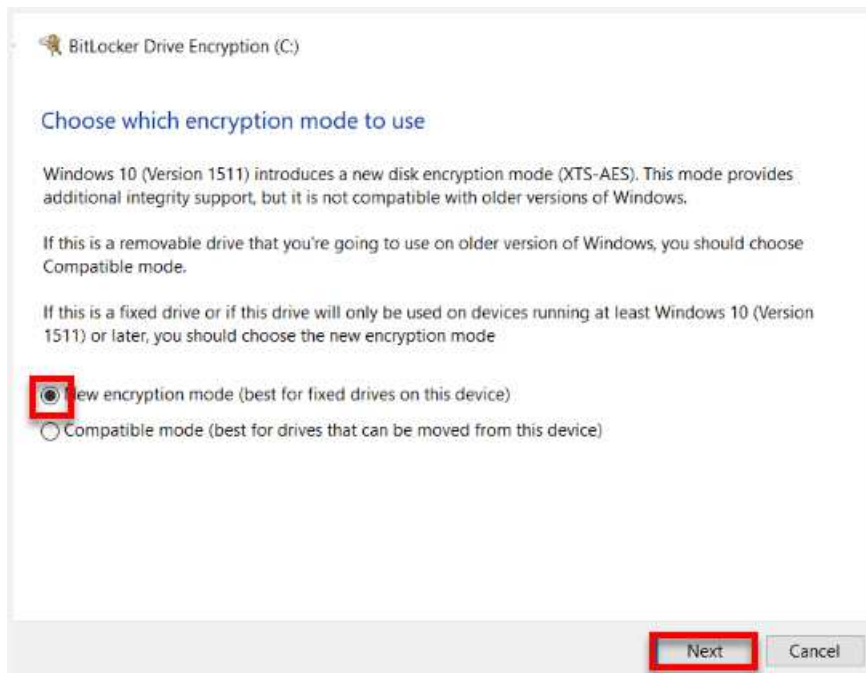
15. Back on the *BitLocker Drive Encryption* window, click **Next**.



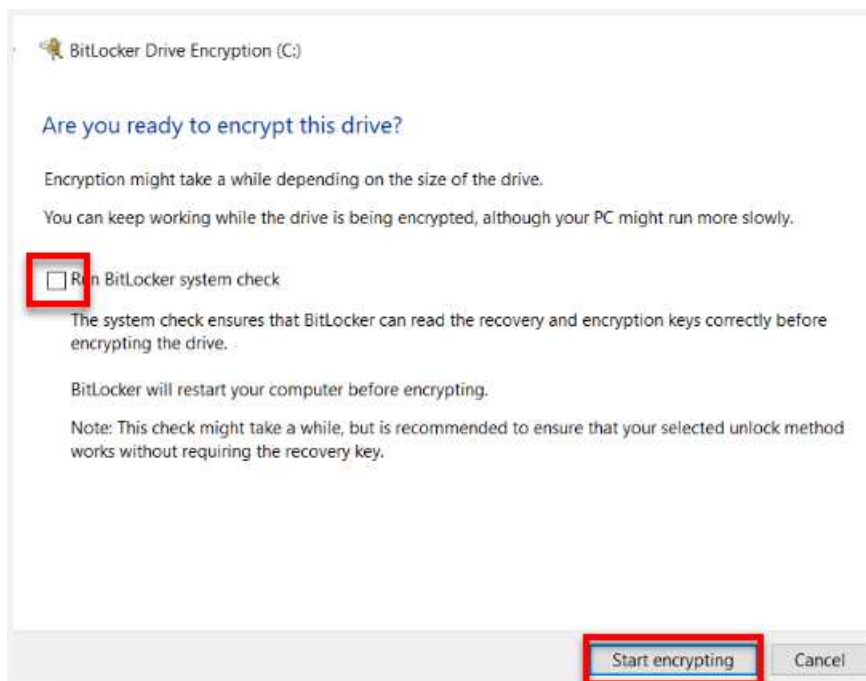
16. Ensure that the radio button for **Encrypt used disk space only** is selected and click **Next**.



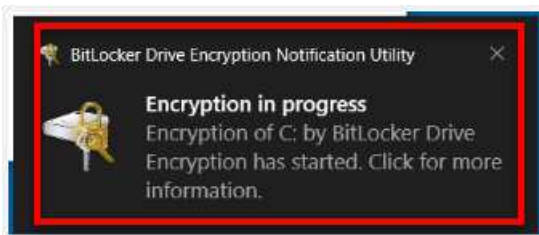
17. On the next step, ensure that the **New encryption mode** is selected and click **Next**.



18. Uncheck **Run BitLocker system check** and click **Start encrypting**.



19. Notice the notification that appears. Click the notification icon to view the encryption progress.

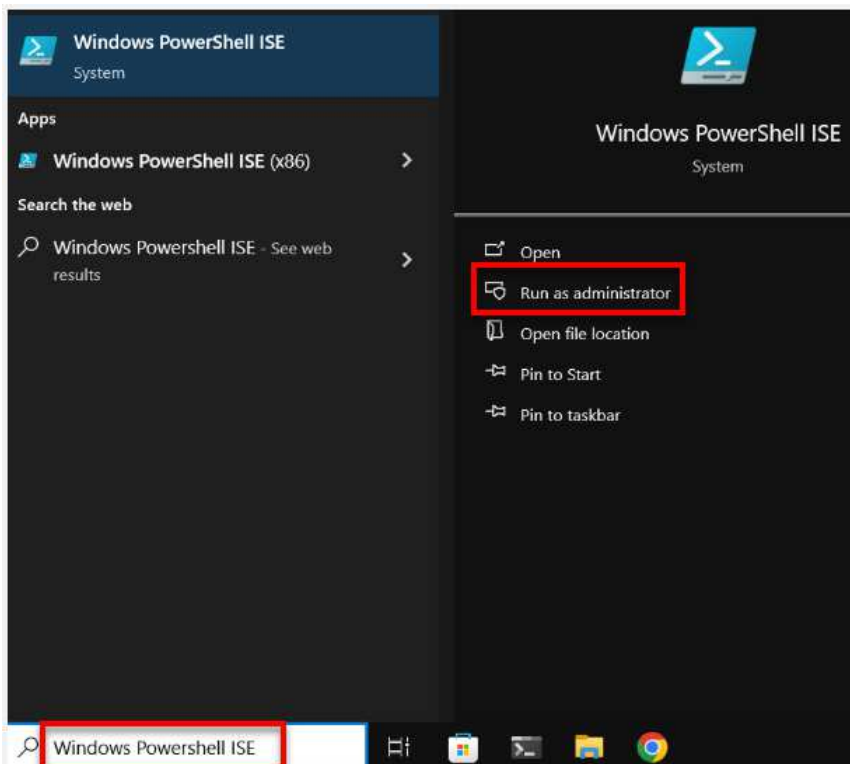


20. The Encryption will take 10 minutes or longer to complete. Please be patient as this process completes. Once the encryption process completes, click **Close** and move on to the next task. Close the *BitLocker Drive Encryption* window.

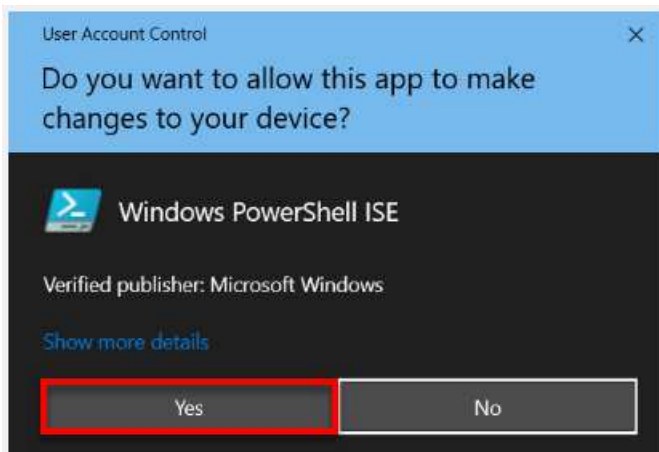


2.3 Disabling BitLocker with PowerShell in Win10-OS

1. In the *Search Box* on the taskbar, type **PowerShell ISE**. Right-click the **Windows PowerShell ISE** entry and select **Run as administrator**.

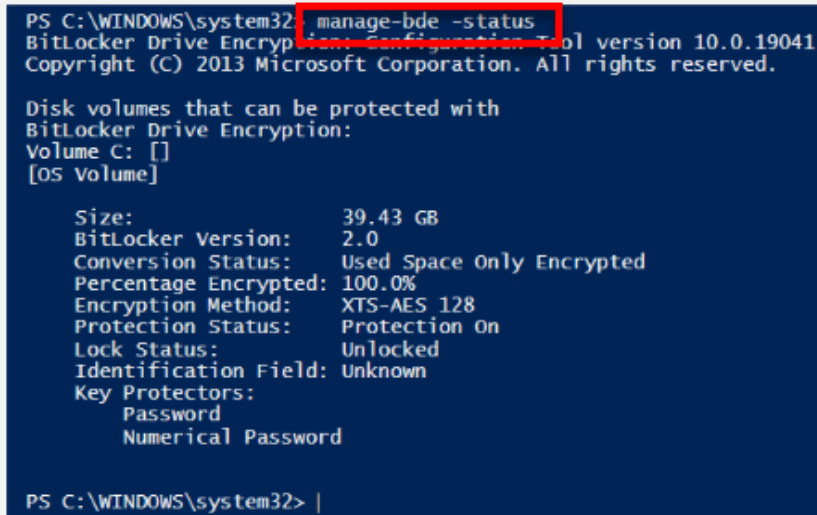


- Click **Yes** to accept the *User Account Control*.



- At the prompt, type `manage-bde -status` followed by pressing the **Enter** key to check the encryption status of each drive.

```
manage-bde -status
```

A screenshot of a Windows command prompt window. The title bar reads "PS C:\WINDOWS\system32". The command prompt shows the command `manage-bde -status` being entered, with the command text highlighted by a red rectangular box. Below the command, the output of the BitLocker Drive Encryption Configuration Tool is displayed. It includes the version (10.0.19041), copyright information, and a list of disk volumes that can be protected. For Volume C: (the OS volume), it shows the size (39.43 GB), BitLocker version (2.0), conversion status (Used Space Only Encrypted), percentage encrypted (100.0%), encryption method (XTS-AES 128), protection status (Protection On), lock status (Unlocked), and identification field (Unknown). It also lists key protectors: Password and Numerical Password. The prompt ends with `PS C:\WINDOWS\system32> |`.

Notice that *Volume C:* is protected and encrypted with *XTS-AES 128*.

- At the prompt, type `Disable-BitLocker -MountPoint "C:"` (include the quotes around "C:") followed by pressing the **Enter** key.

```
Disable-BitLocker -MountPoint "C:"
```

```
PS C:\WINDOWS\system32> Disable-Bitlocker -MountPoint "C:"

ComputerName: WIN10-05

VolumeType      Mount Point CapacityGB VolumeStatus      Encryption KeyProtector
-----
OperatingSystem C:          39.43 DecryptionInProgress 59 {Password, Reco...
```

- Remove the encryption from the C: drive by entering the `manage-bde -off C:` command.

```
manage-bde -off C:
```

```
PS C:\WINDOWS\system32> manage-bde -off C:
BitLocker Drive Encryption Configuration Tool version 10.0.19041
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

Decryption is now in progress.

PS C:\WINDOWS\system32> |
```



The decryption continues in the background until the drive is in a fully decrypted status.

6. Enter the `manage-bde -status` command to check the status of decryption progress while it is running in the background.

```
manage-bde -status
```

```
PS C:\WINDOWS\system32> manage-bde -status
BitLocker Drive Encryption: Configuration Tool version 10.0.17134
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

Disk volumes that can be protected with
BitLocker Drive Encryption:
Volume C: [ ]
[OS Volume]

Size: 39.51 GB
BitLocker Version: 2.0
Conversion Status: Decryption in Progress
Percentage Encrypted: 24.1%
Encryption Method: XTS-AES 128
Protection Status: Protection Off
Lock Status: Unlocked
Identification Field: Unknown
Key Protectors:
    Password
    Numerical Password

PS C:\WINDOWS\system32>
```

7. Once the drive is completely decrypted, type `exit` followed by pressing the **Enter** key to close *PowerShell*.

```
PS C:\WINDOWS\system32> manage-bde -status
BitLocker Drive Encryption: Configuration Tool version 10.0.17134
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

Disk volumes that can be protected with
BitLocker Drive Encryption:
Volume C: [ ]
[OS Volume]

Size: 39.51 GB
BitLocker Version: None
Conversion Status: Fully Decrypted
Percentage Encrypted: 0.0%
Encryption Method: None
Protection Status: Protection Off
Lock Status: Unlocked
Identification Field: None
Key Protectors: None Found

PS C:\WINDOWS\system32> exit
```

8. At this point, you have completed the *BitLocker* procedure; you may end the reservation.